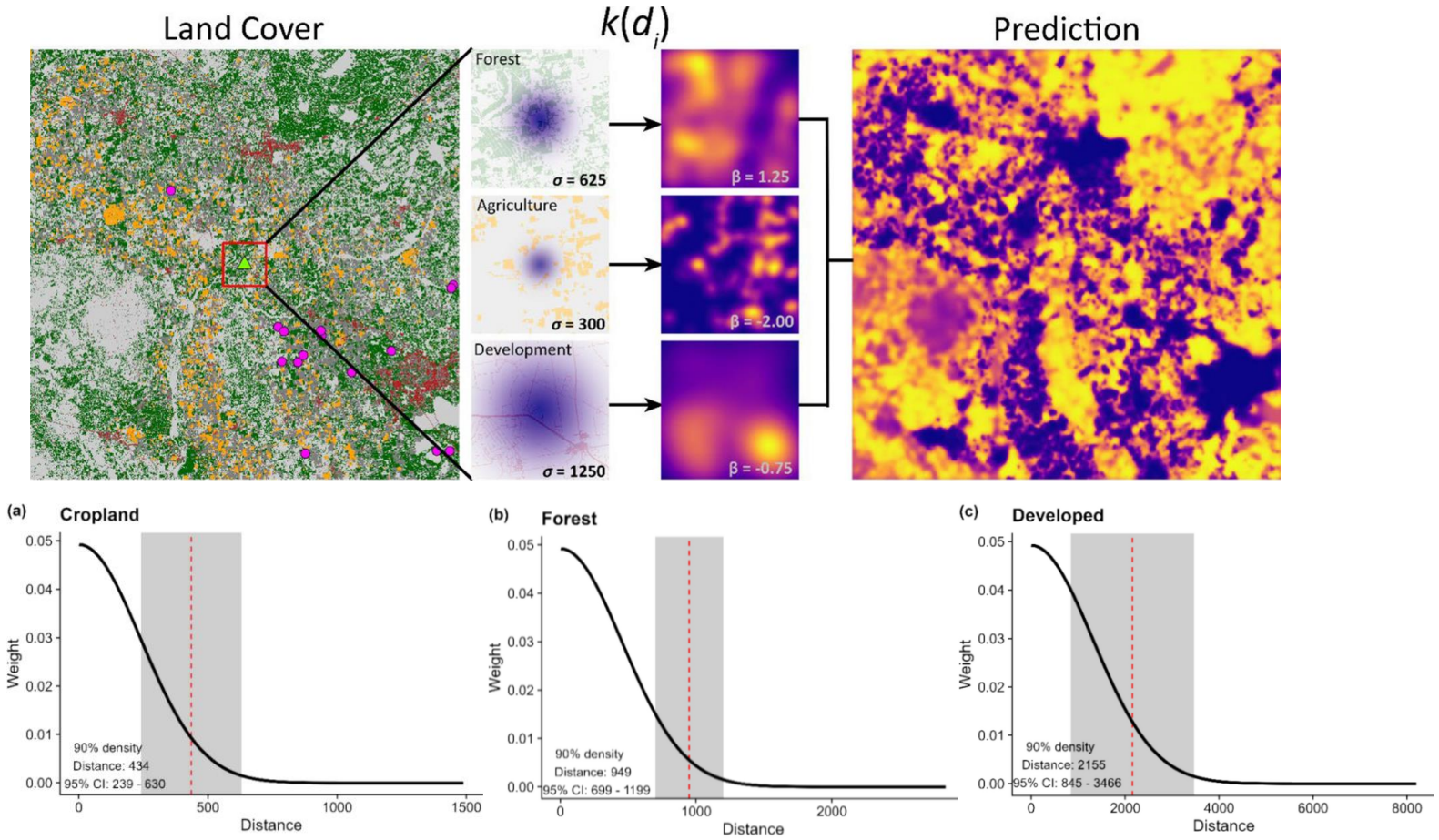
multiScaleR

Approachable and generalizable multiscale ecological modeling in R

Bill Peterman & Joe Drake

Terrestrial Wildlife Ecology Lab,
Ohio State University

Workshop for IALE-North America Annual Meeting, April 2026, Athens, GA



1 Some background

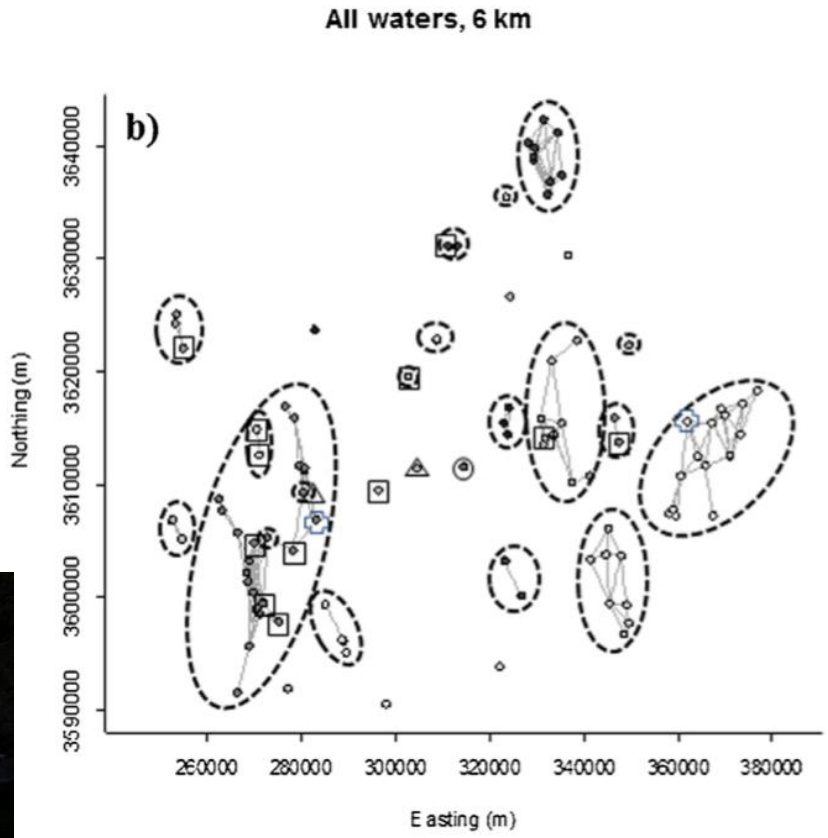
The importance of space for ecology

- Ecology
  - Resource availability and use
  - Space use within and throughout geographical range
  - Movement, dispersal, and migration
  - Inter- and intra-species interactions





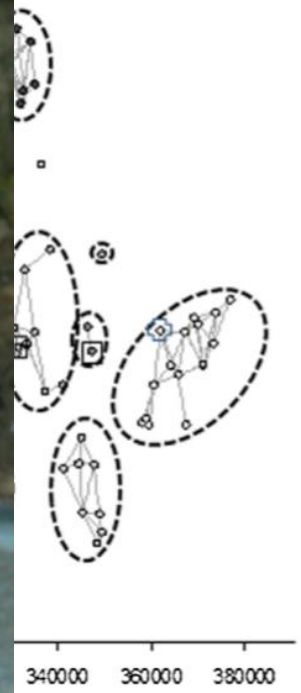




Fletcher & Fortin 2018; McIntyre et al. 2016

- Ecolog
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  - Inte

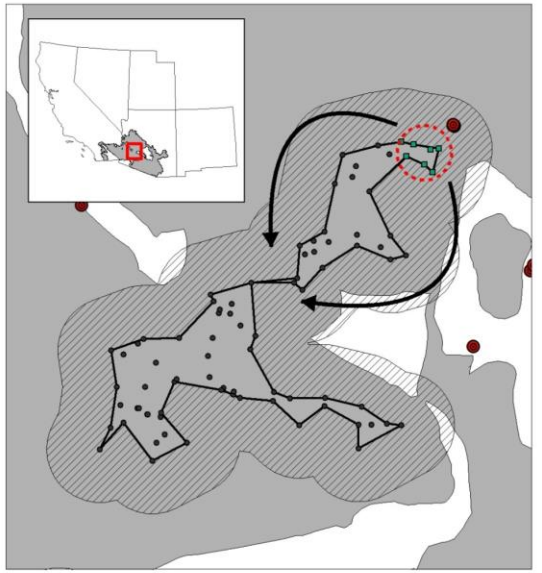


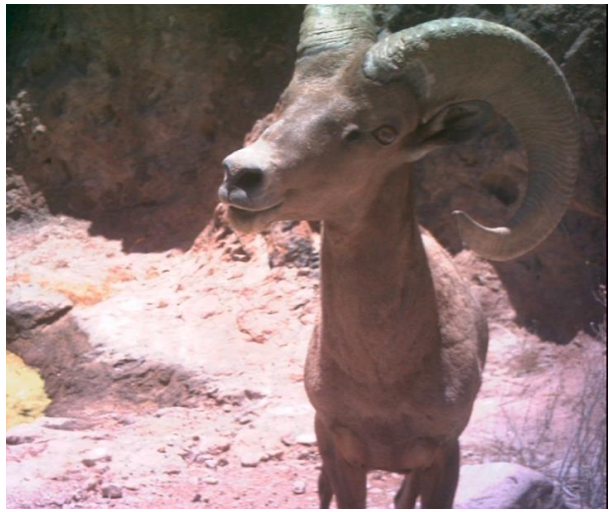


ntyre et al. 2016

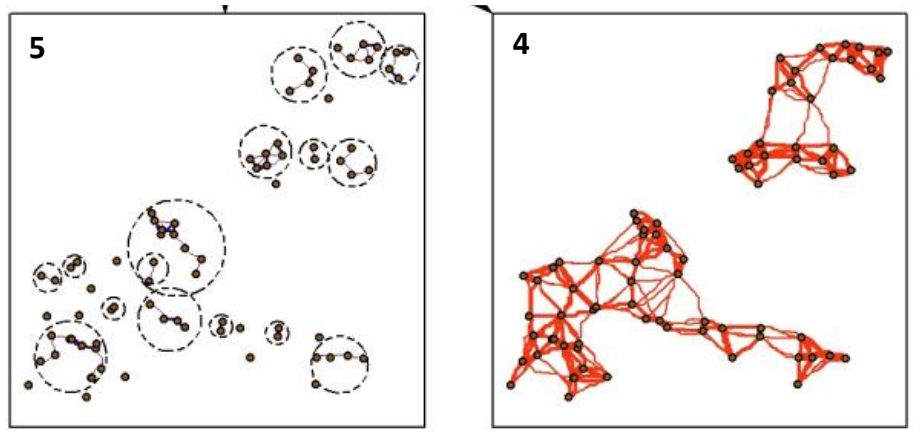
The importance of space for conservation

- Conservation
  - Spatial mapping of biodiversity and ecosystem services
  - Informs mitigation efforts on effects of environmental change
  - Helps improve prioritization efficacy for areas for conservation
  - Critical component of tools and models used in conservation



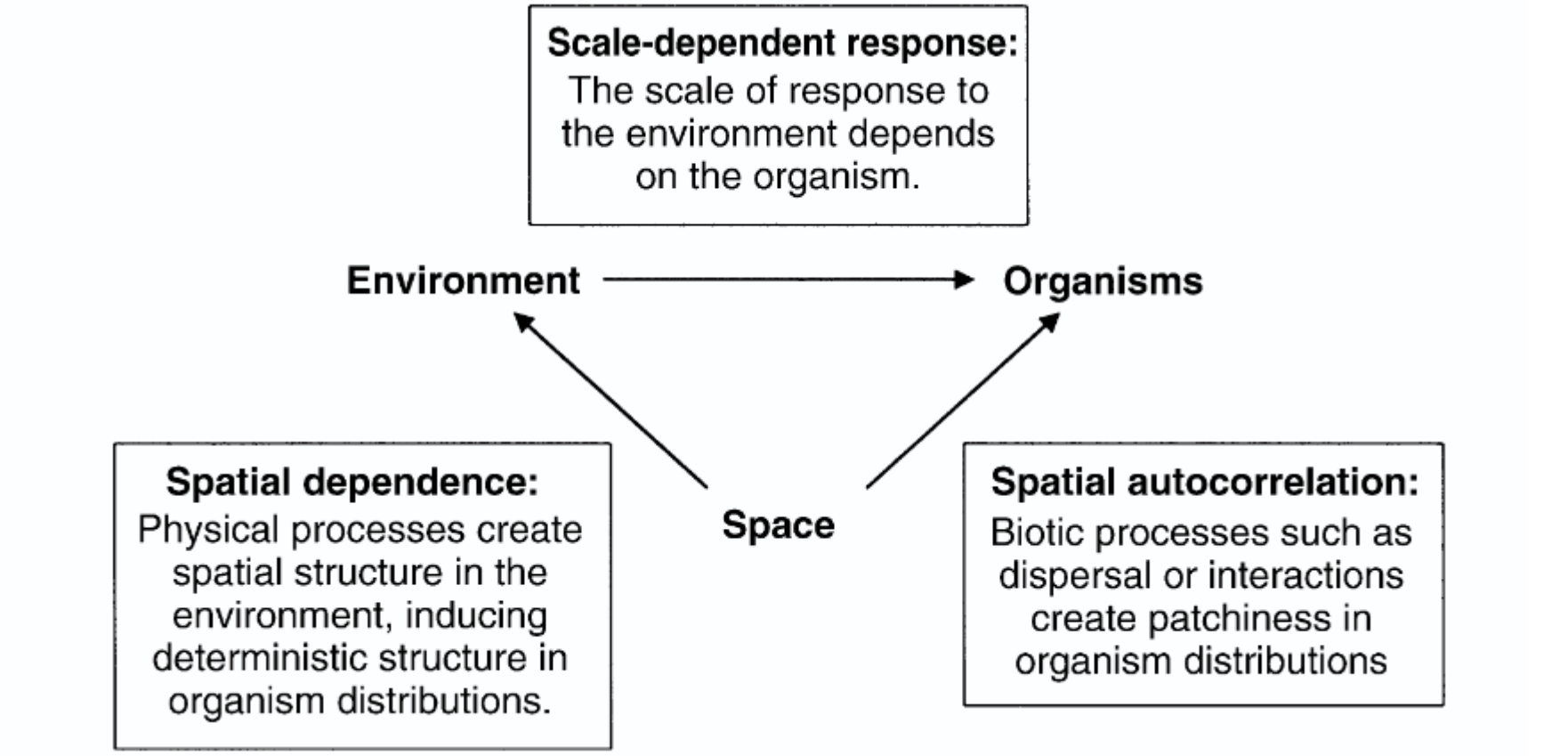






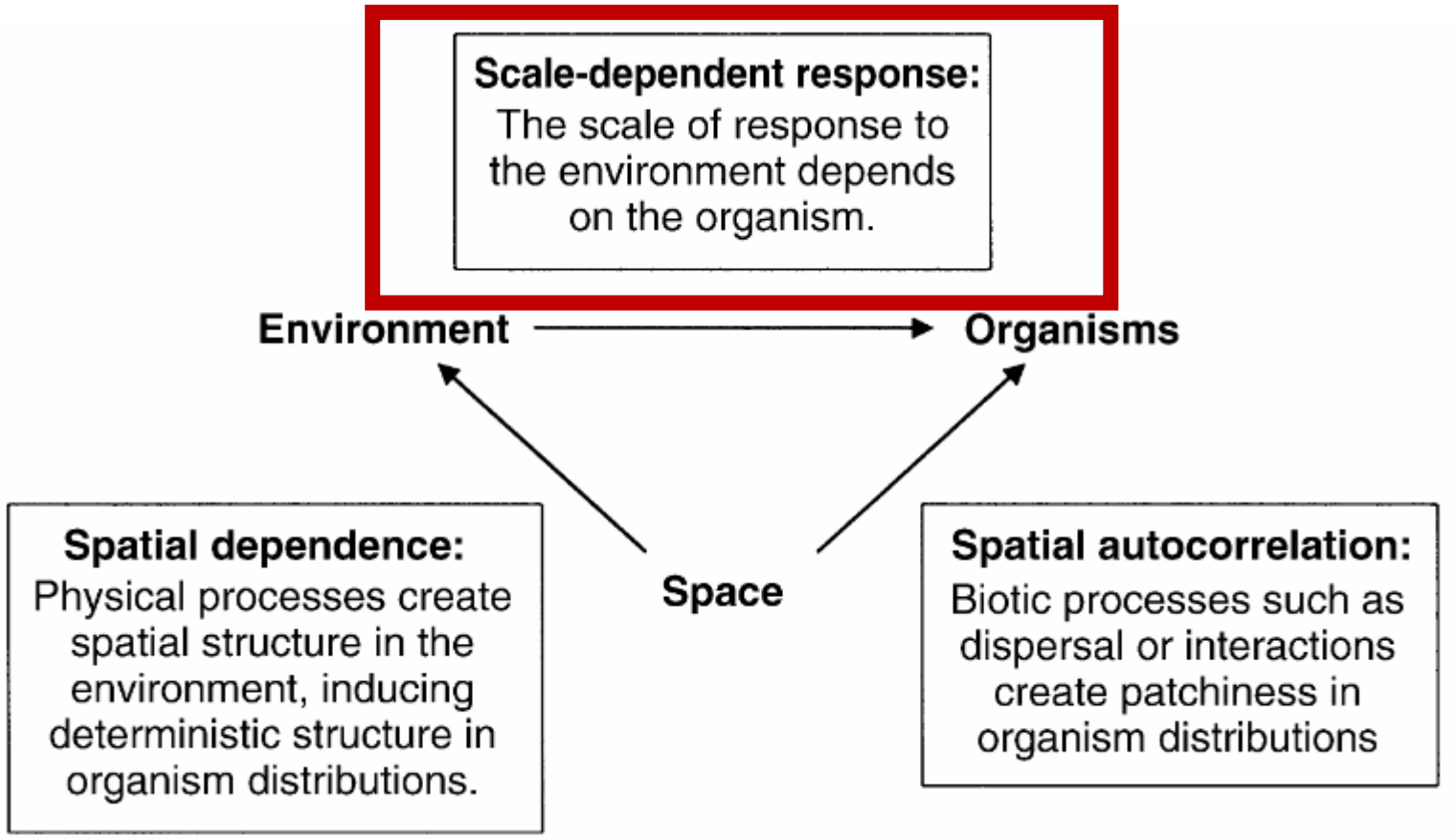
Fletcher & Fortin 2018; Drake et al. 2017

Spatial effects in ecological data



Wagner and Fortin 2005

Spatial effects in ecological data



Wagner and Fortin 2005

What is scale?

- The dimension or domain of an object, process, or pattern across space or time.

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  - Spatial
    - Grain = smallest unit of measure (e.g. raster cell size)
    - Extent = size of the study area (e.g. raster bounding box)
  - Temporal
    - Temporal resolution = frequency of sampling
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Intrinsic vs. Imposed Scale

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  - Independent of observation/measurement
  - Inherent to the phenomenon: object, pattern, or process
  - A red-spotted toadlet sees the landscape very differently than a Harris hawk





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  - Experimental scale, management scale, analysis scale, policy scale, etc….

Wu and Li 2006; Millington 2022

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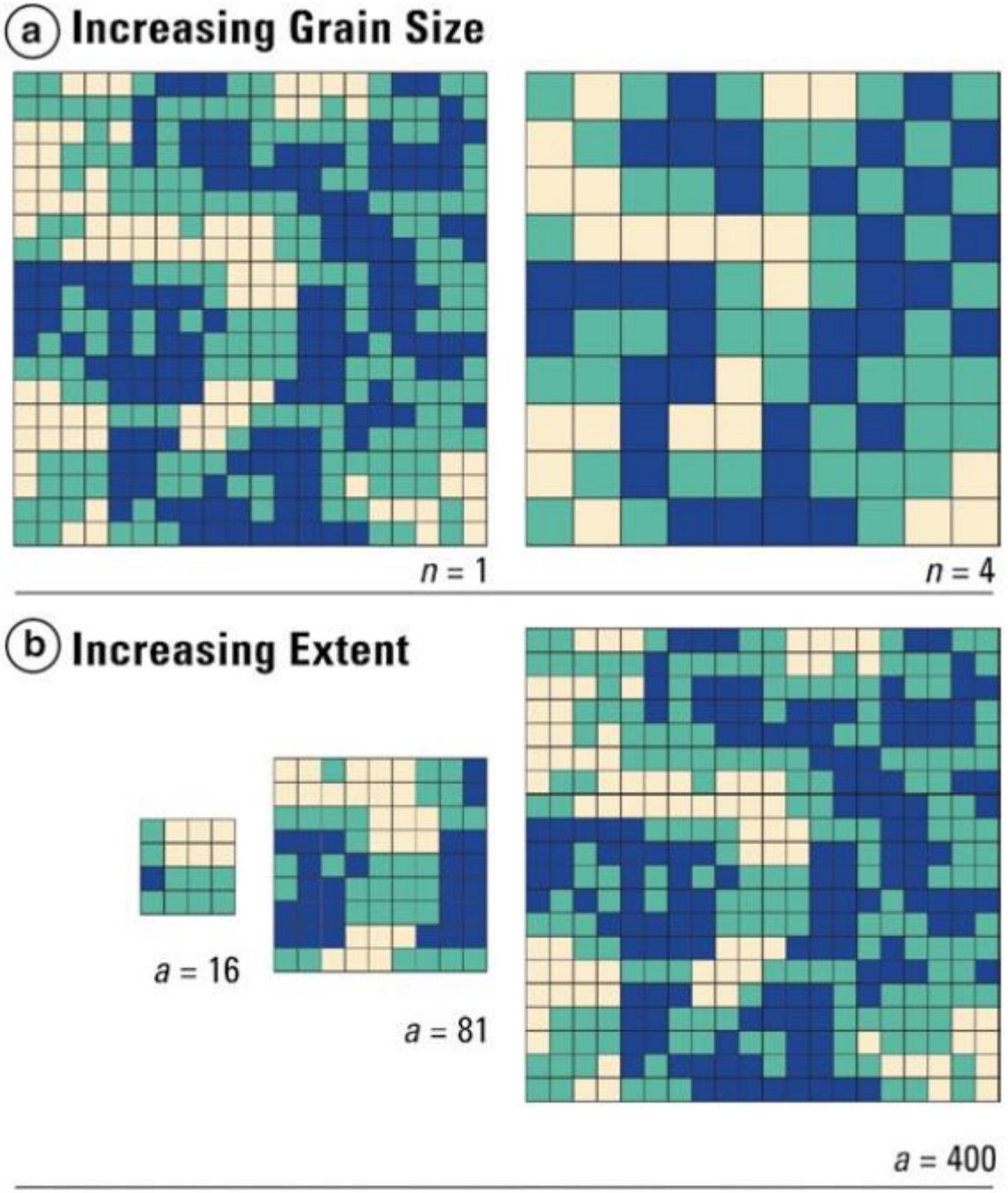
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Wu and Li 2006; Millington 2022

Grain and Extent

- As cells are aggregated, the grain size increases, coarsening the resolution and smoothing the data

- As the extent increases, more of the “landscape” is observed; additional patterns or processes may occur as extent increases

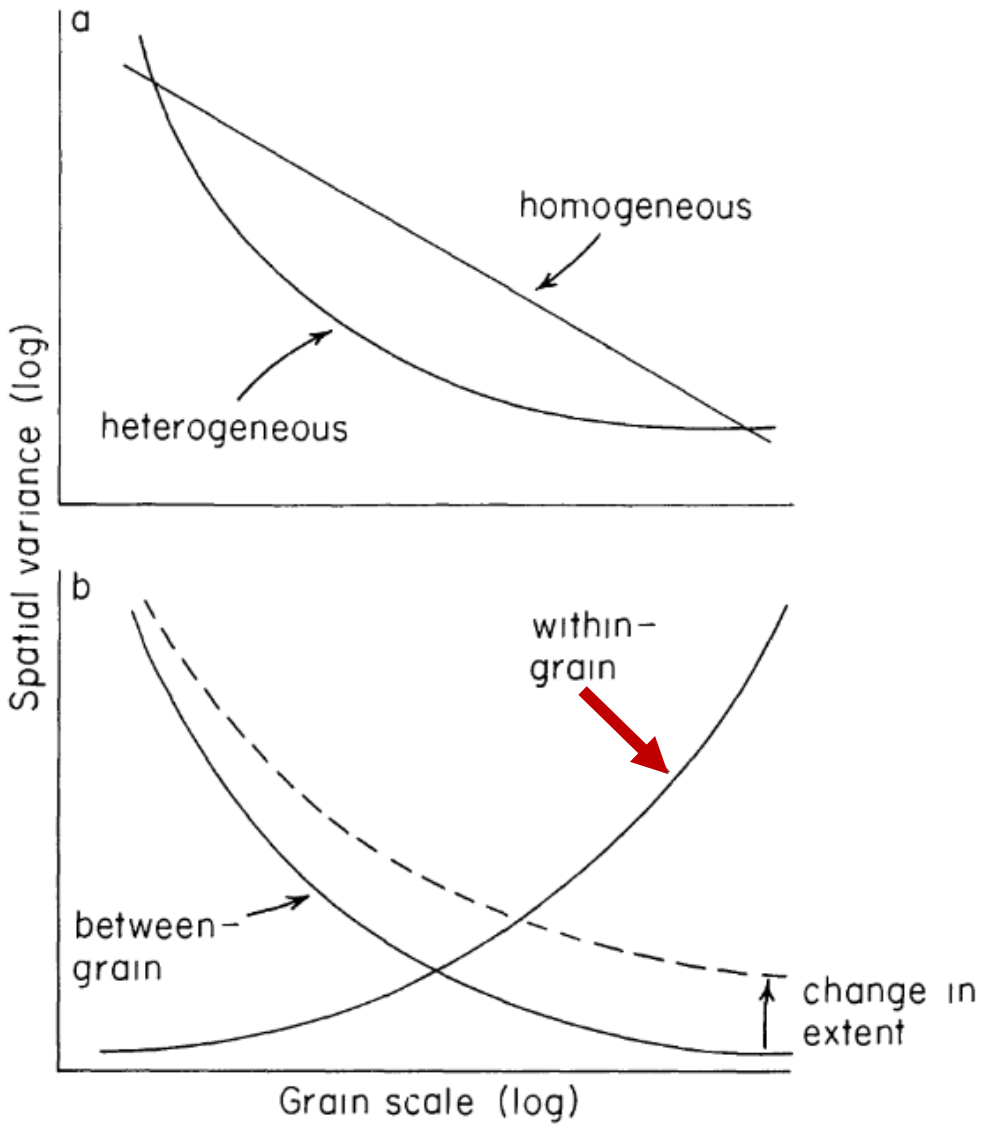


Tuner & Gardner 2015

Grain and Extent

As grain increases

- Greater proportion of heterogeneity contained within a sample/unit/cell
  - Loss of resolution
- Variance between grains reduces
  - Rare species less likely to be recorded (diversity decreasing) if using habitat threshold requirements
  - Rare species occurrence more likely to be recorded (diversity increasing)

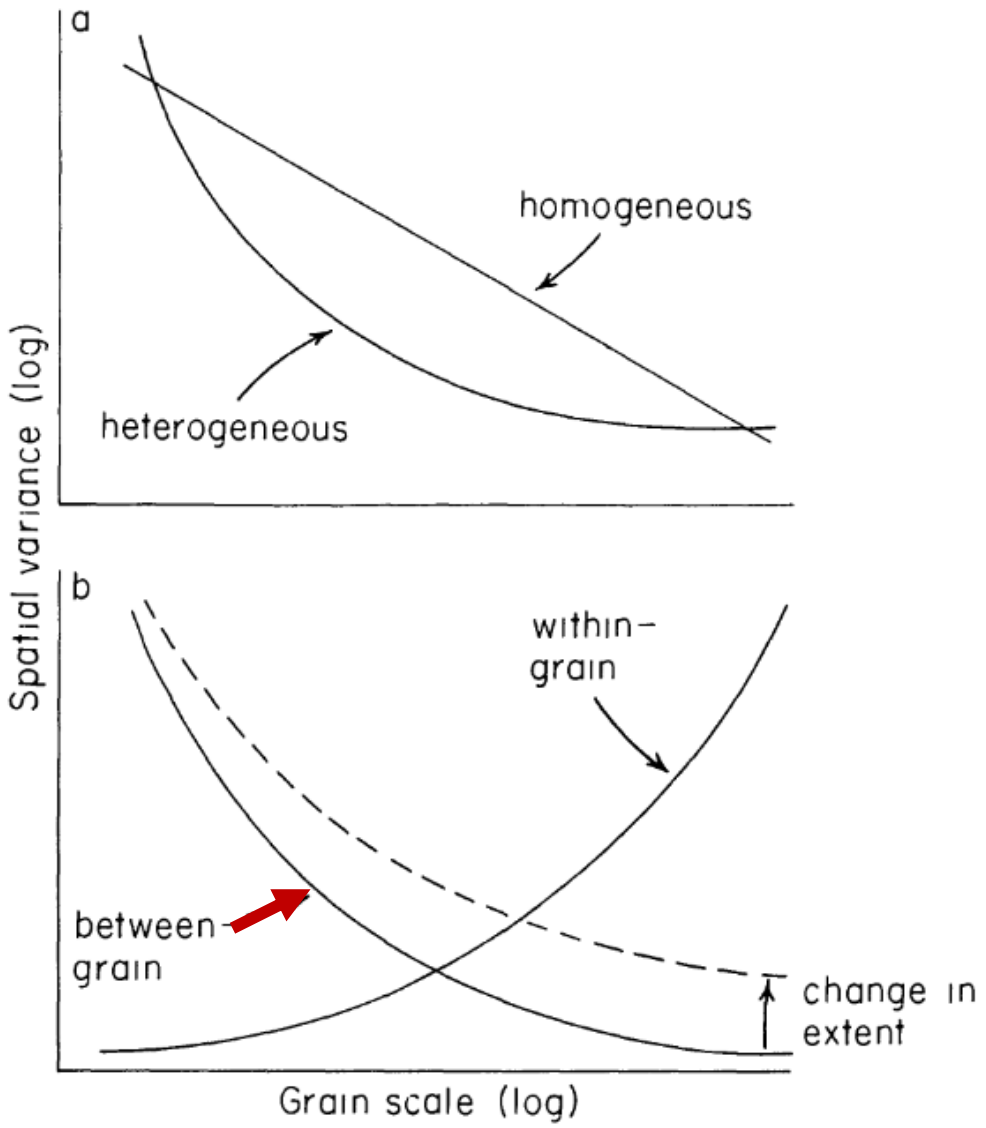


Wiens 1989

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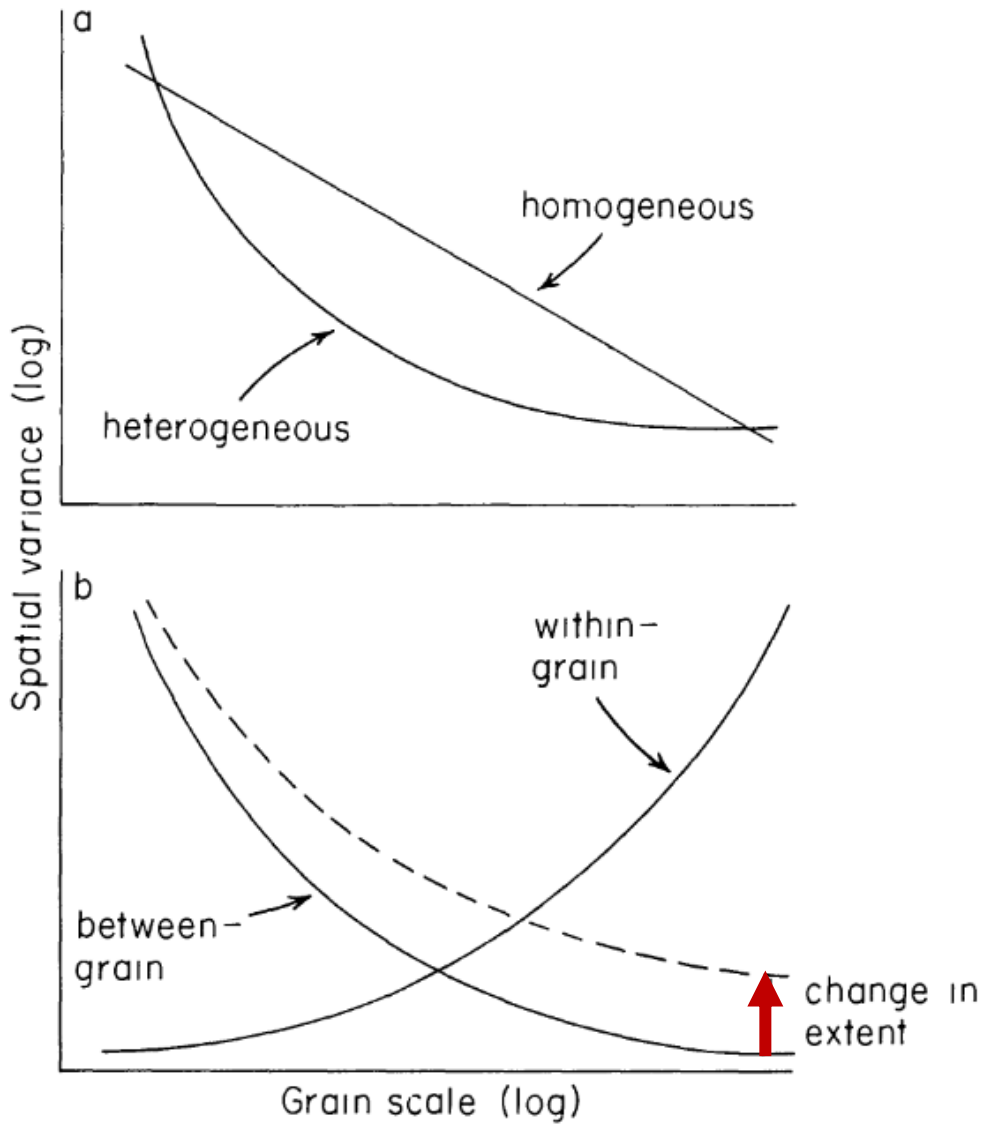


Wiens 1989

Grain and Extent

As extent increases

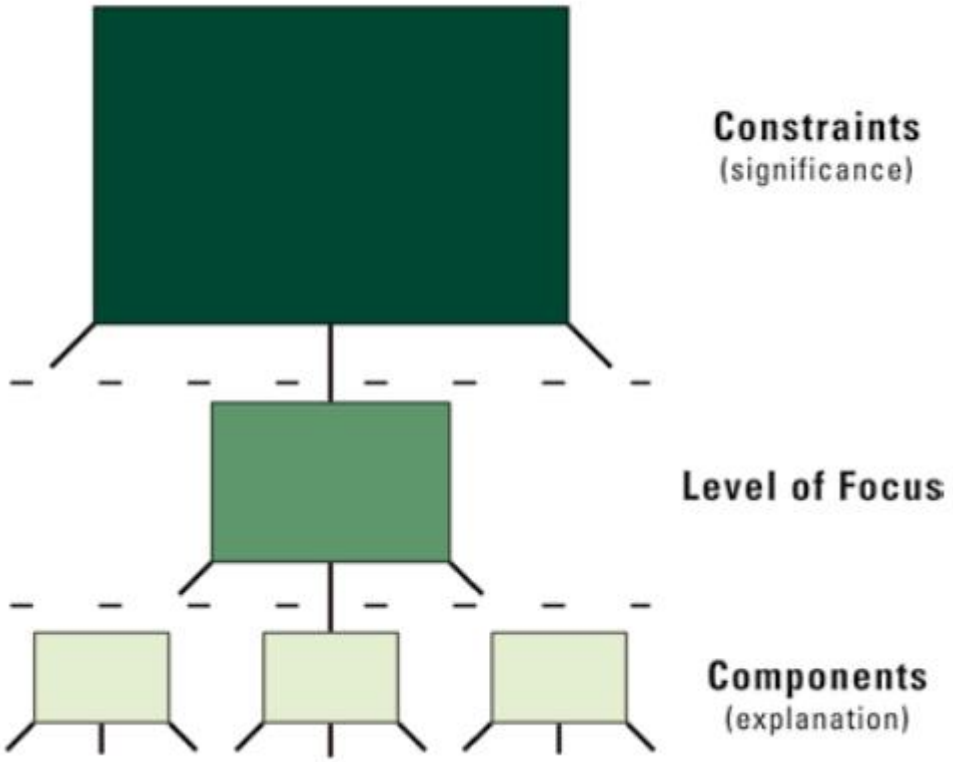
- Spatial heterogeneity changes
- Variance between grains increases (generally)



Wiens 1989

Scale vs. Hierarchy vs. Levels of Organization

- Hierarchy describes interconnected levels that may influence the levels above (+1) or constrain the levels below (-1) the focal level (0).



O'Neill et al. 1986

Scale vs. Hierarchy vs. Levels of Organization

- Hierarchy describes interconnected levels that may influence the levels above (+1) or constrain the levels below (-1) the focal level (0).
  - The reach system (0) is constrained within the stream or segment (+1)
  - The pools (-1) within the reach (0) may explained by the details of the substrate.

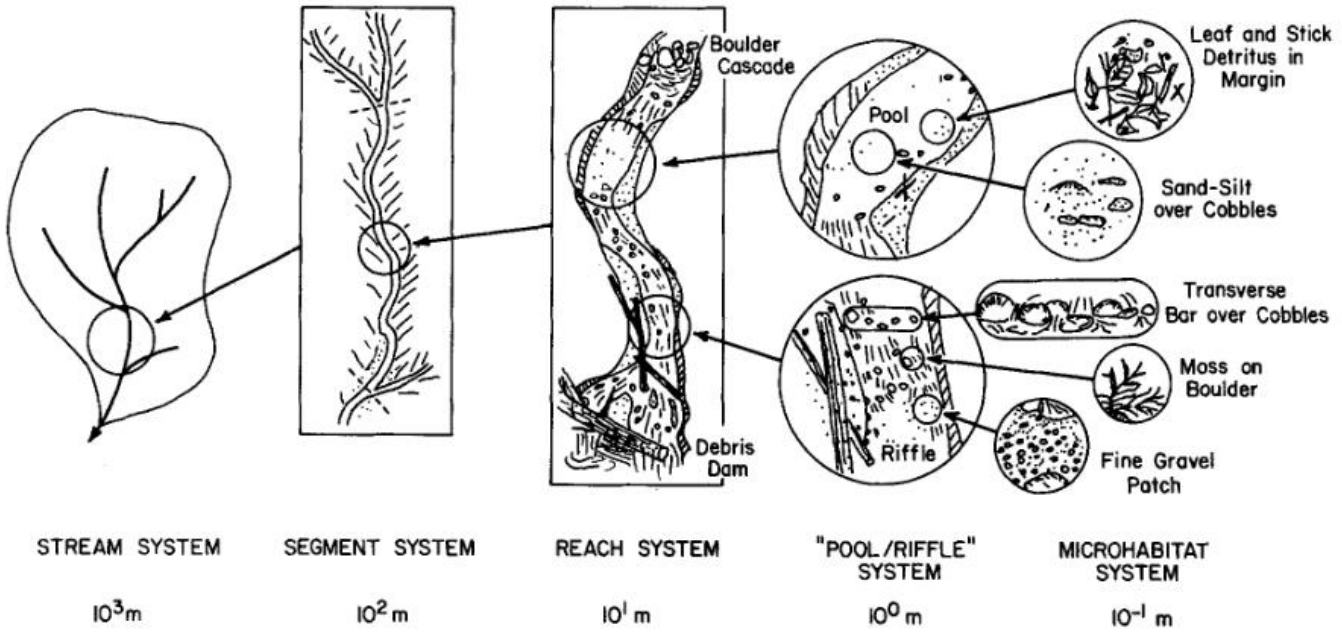
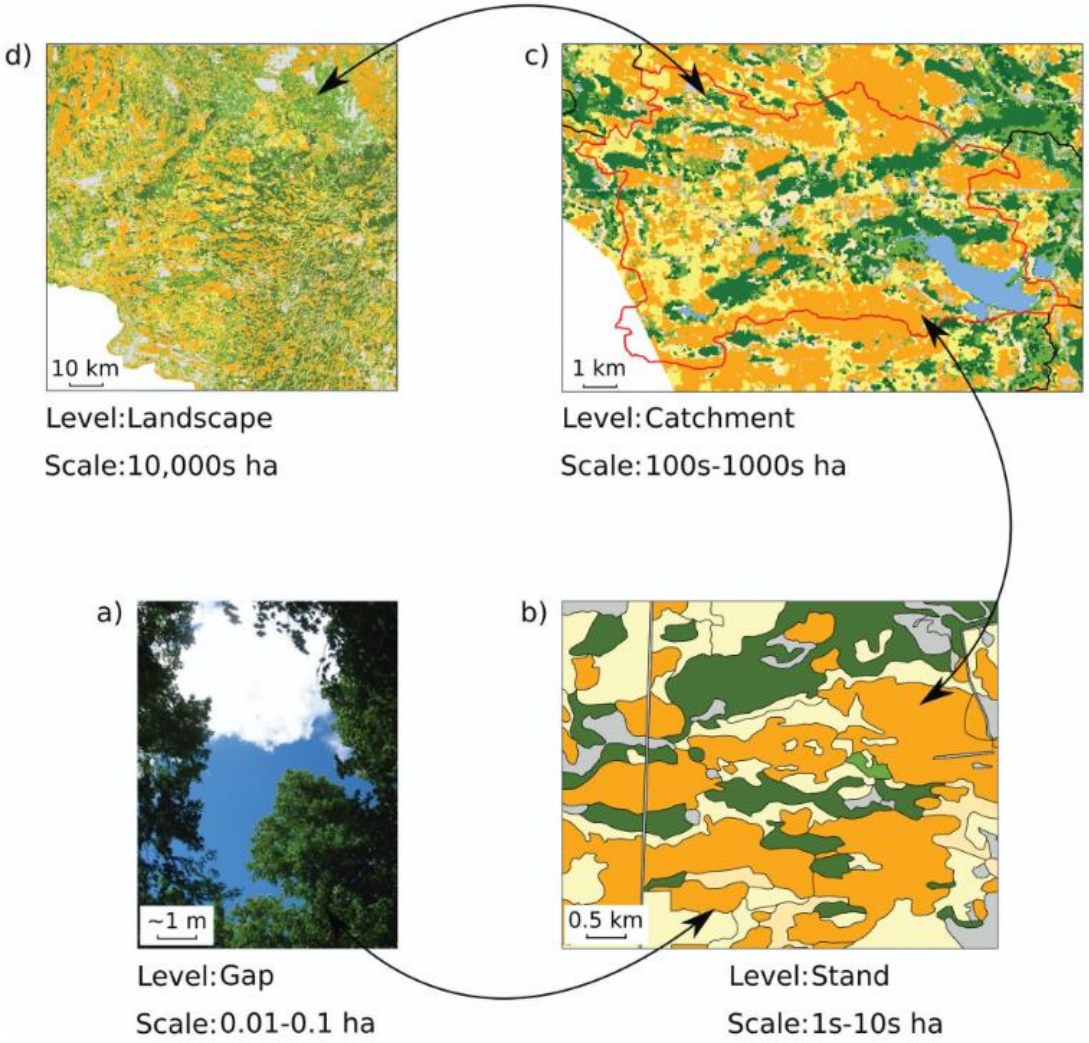


Figure 2. Hierarchical organization of a stream system and its habitat subsystems. Approximate linear spatial scale, appropriate to second- or third-order mountain stream, is indicated.

Frissell et al. 1986

Scale vs. Hierarchy vs. Levels of Organization

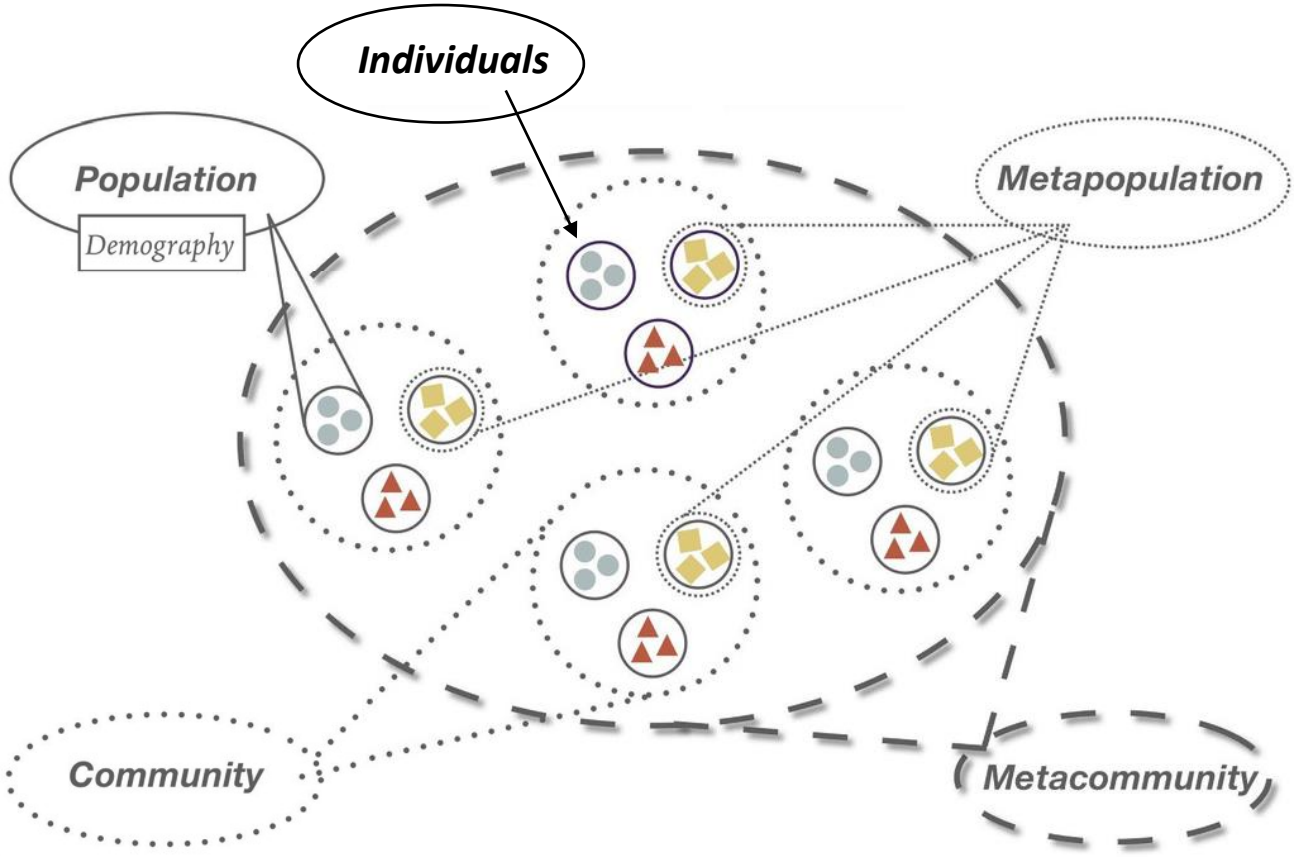
- Landscapes can be interpreted as a hierarchical mosaic of patches.
- Scale is distinct from these “levels” or suborganizations of larger patterns/processes
- Hierarchical perspectives can help inform the scale to investigate or observe/measure a process



Urban et al. 1987; Millington 2022

Scale vs. Hierarchy vs. Levels of Organization

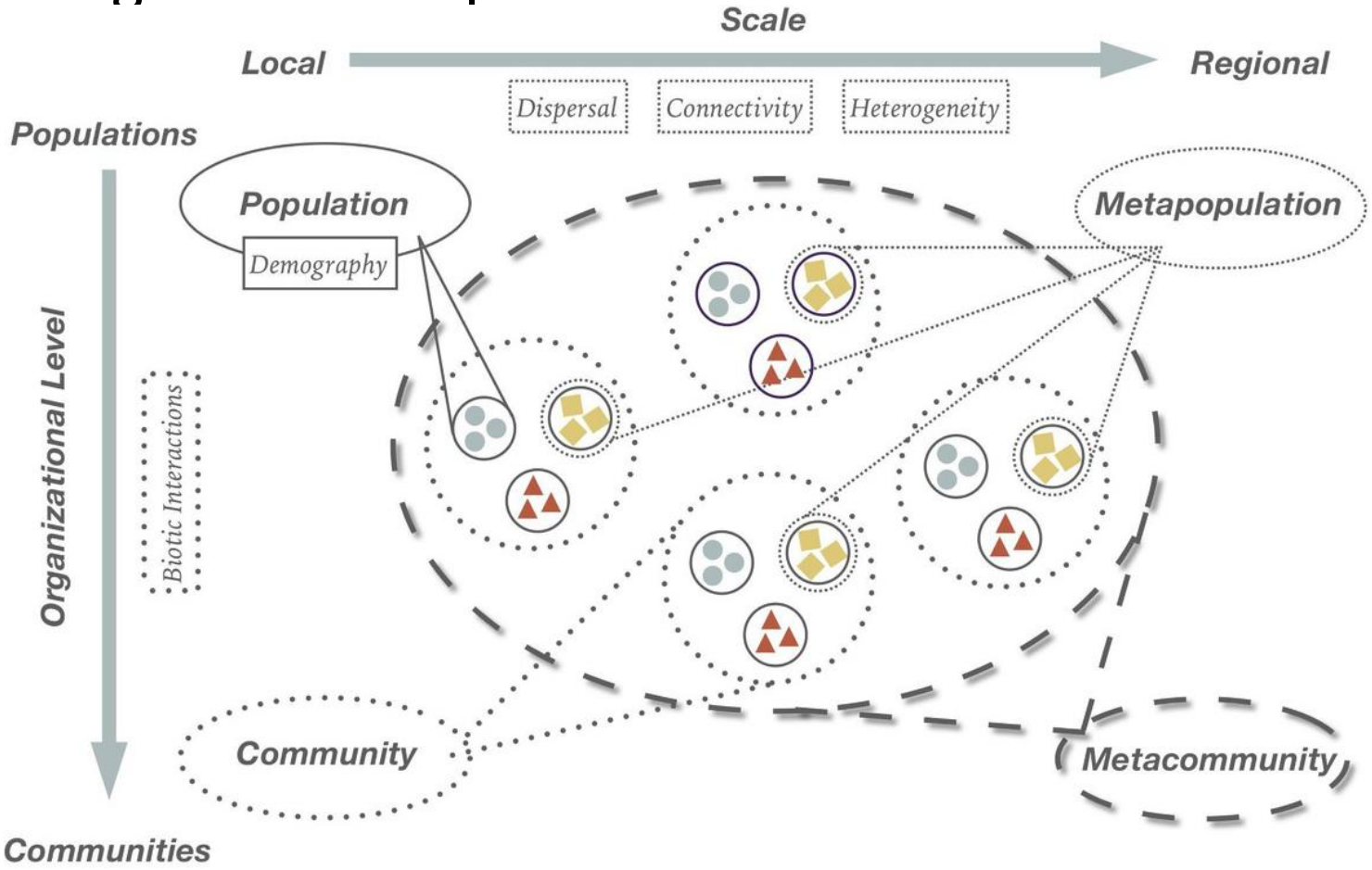
- Levels of organization – place within biotic structures and hierarchy



Chase et al. 2020

Scale vs. Hierarchy vs. Levels of Organization

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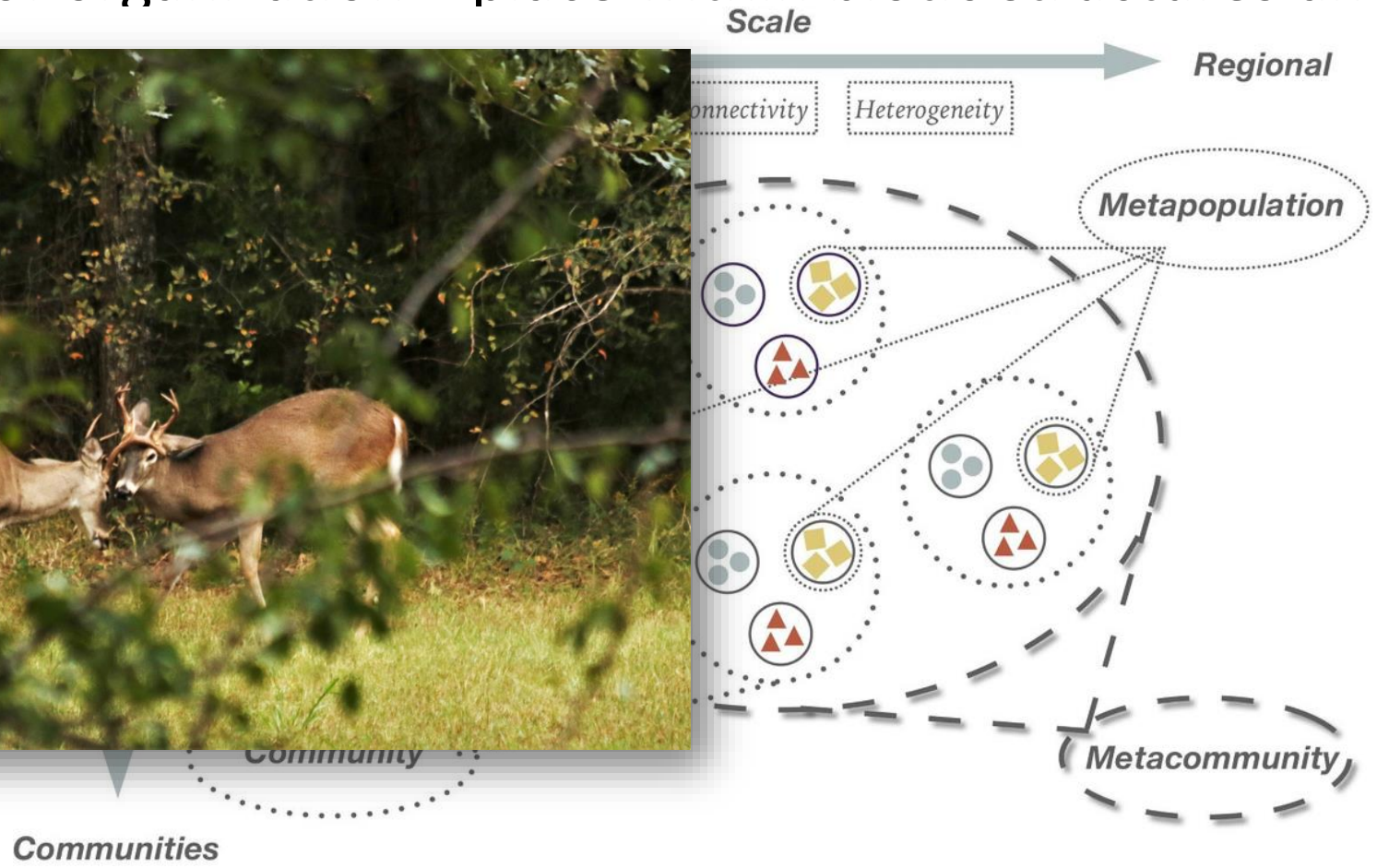


Chase et al. 2020

Scale vs. Hierarchy vs. Levels of Organization

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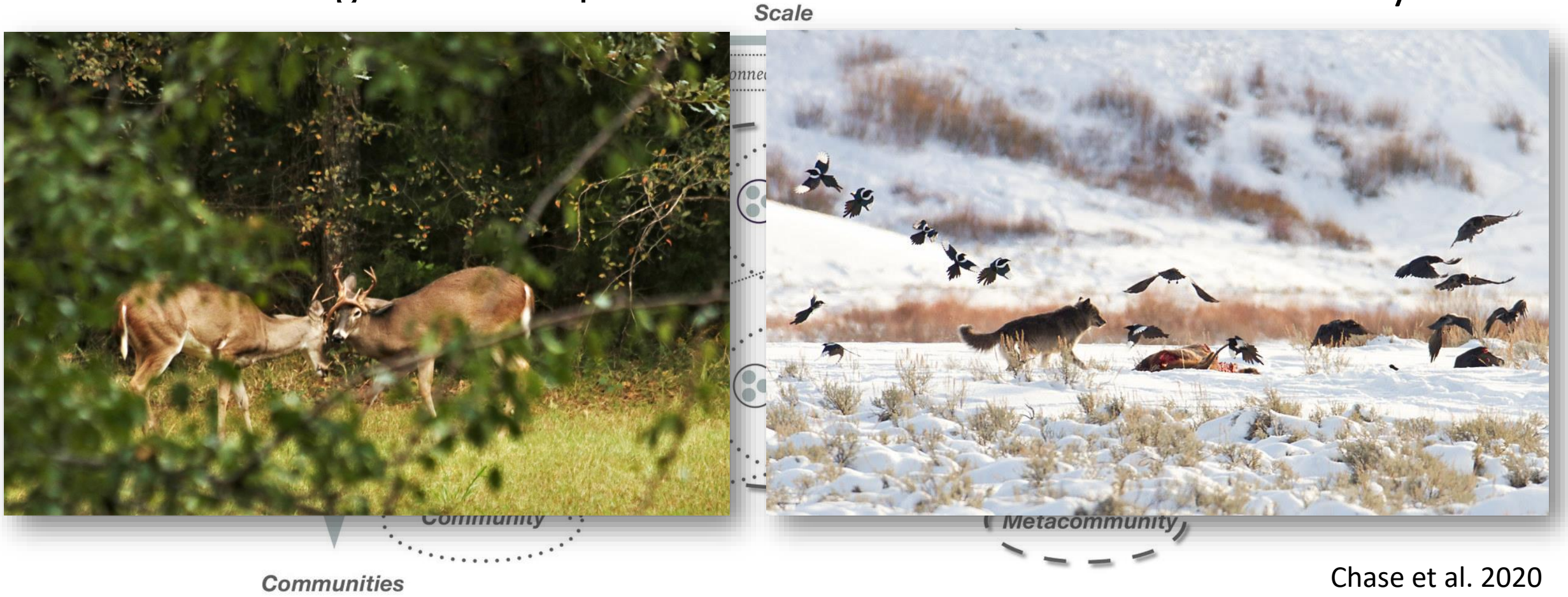




Chase et al. 2020

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Chase et al. 2020

Why is it important?

Scale may impact…

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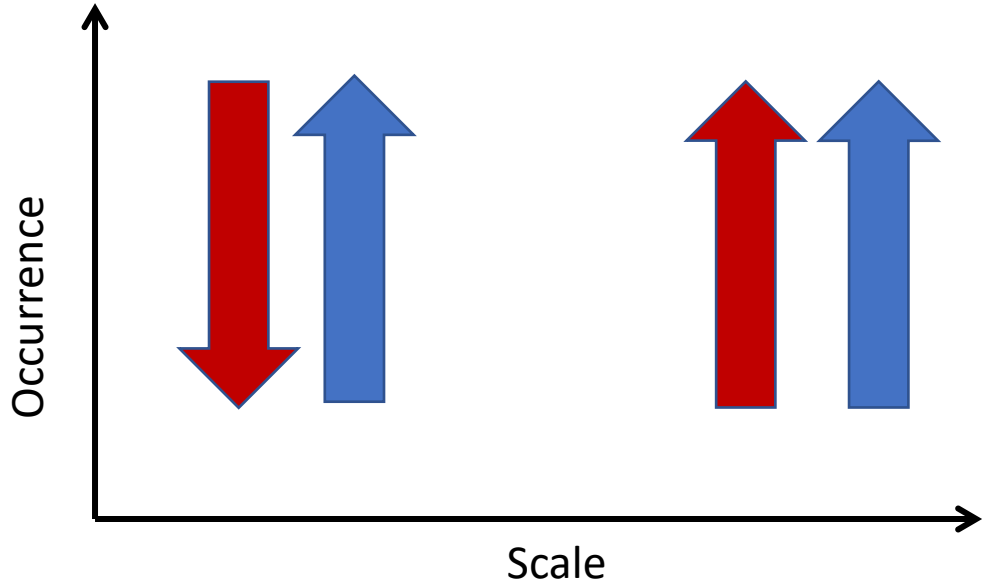
1. The role of biotic and abiotic interactions
2. How “open” or “closed” a system is
3. Relationships of ecological patterns and processes
4. Efficacy of conservation and management actions

American redstarts



least flycatchers



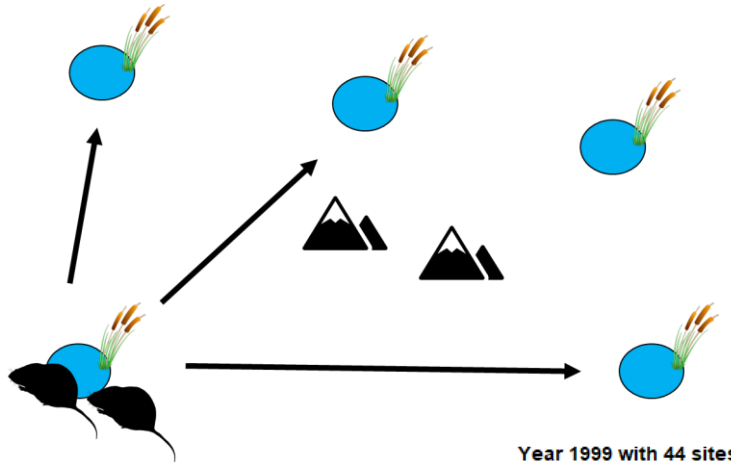


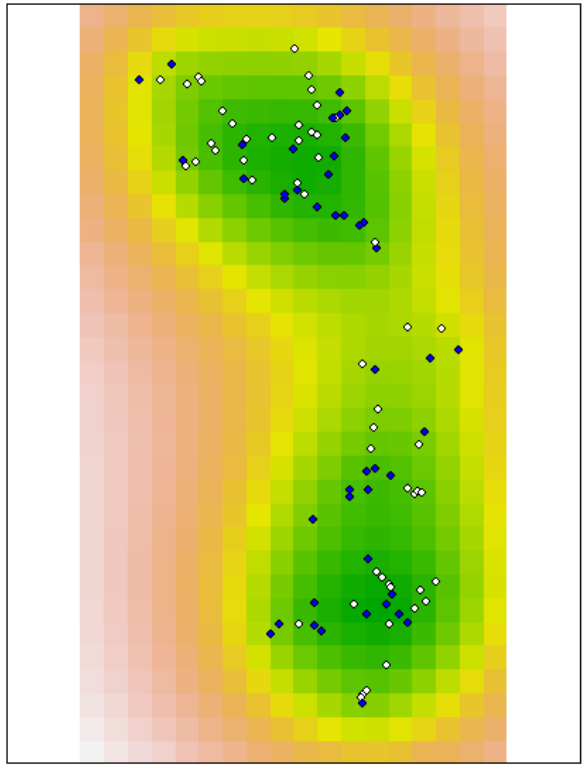
Sherry and Holmes 1988; Fletcher 2007

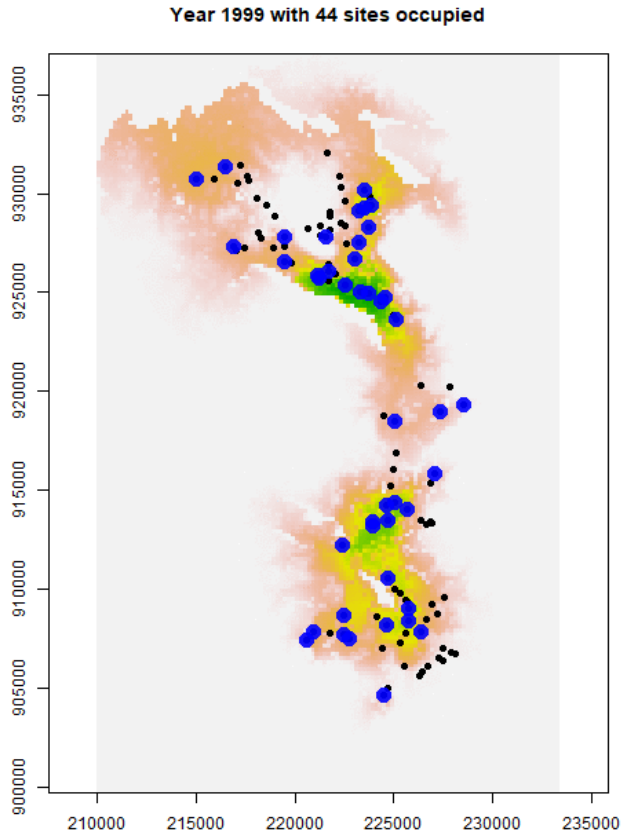
Why is it important?

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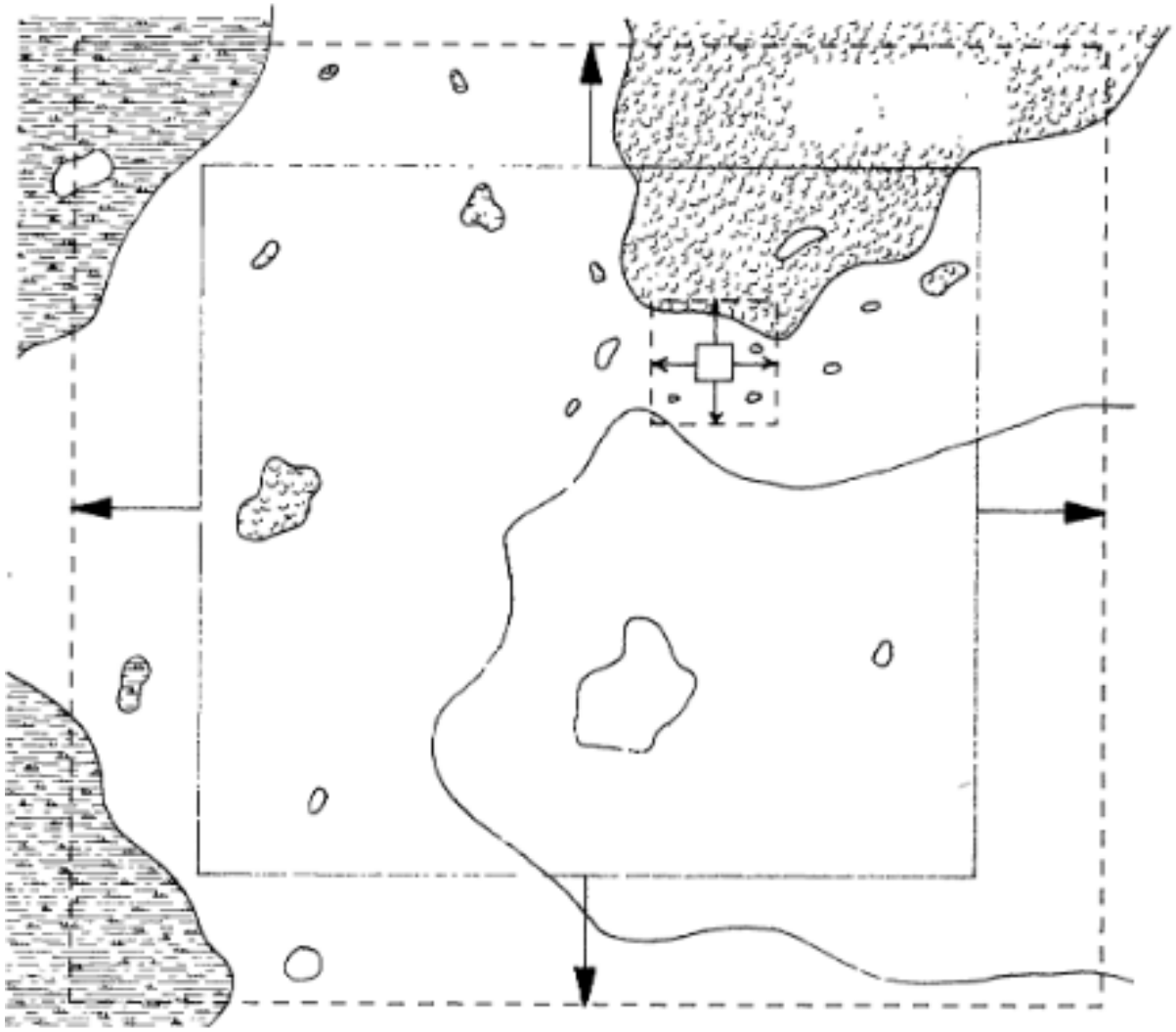


Drake 2021

Why is it important?

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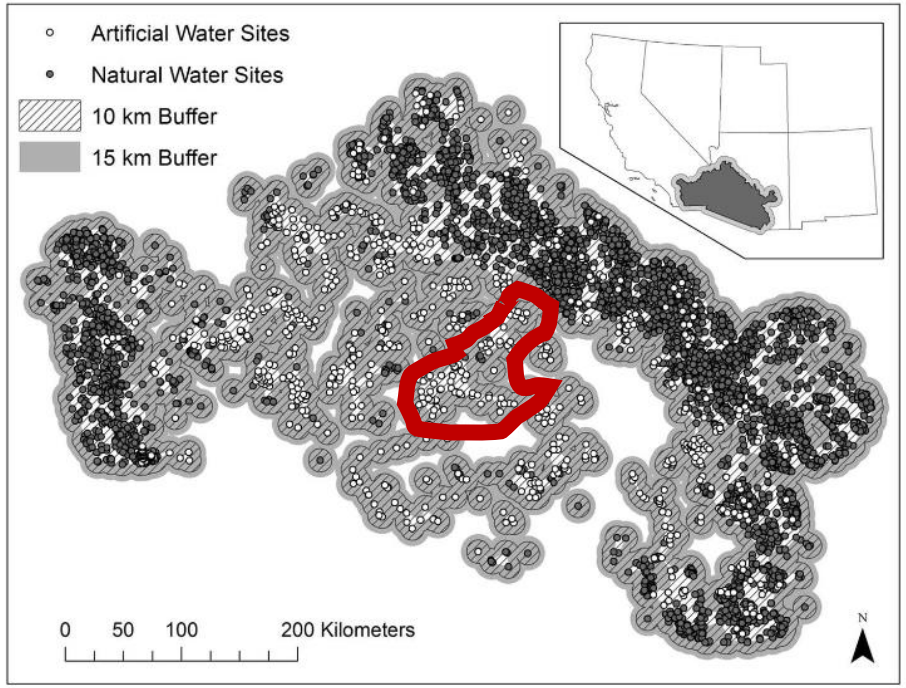


Wiens 1989

Why is it important?

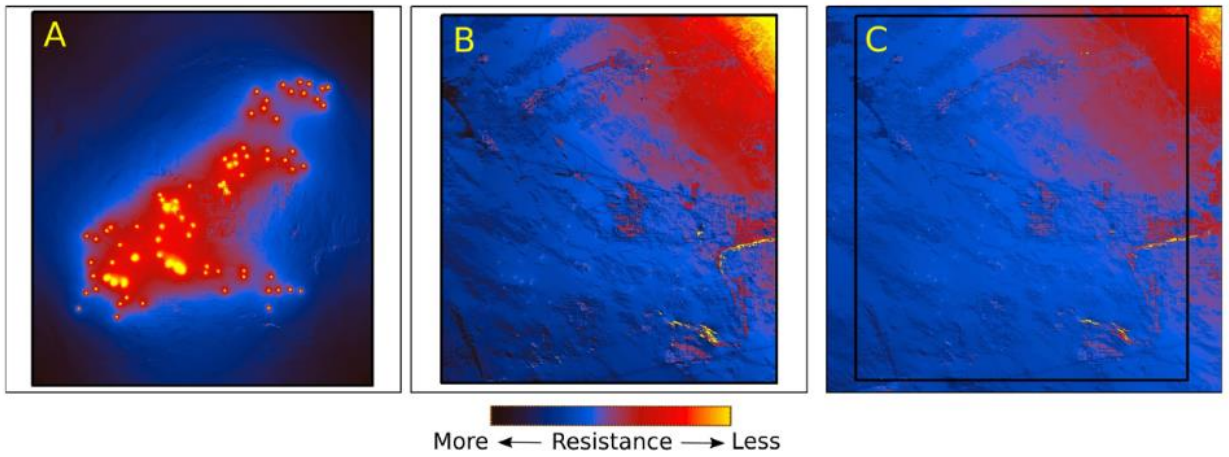
Scale may impact…

1. The role of biotic and abiotic interactions
2. How “open” or “closed” a system is
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    1. dispersal vs. connectivity
    2. uniqueness of species assembly



Local Connectivity

Regional Connectivity



Drake et al. 2017a,b

What drives the (intrinsic) scale of effect?

Landscape Ecol (2016) 31:1177–1194
DOI 10.1007/s10980-015-0314-1

RESEARCH ARTICLE

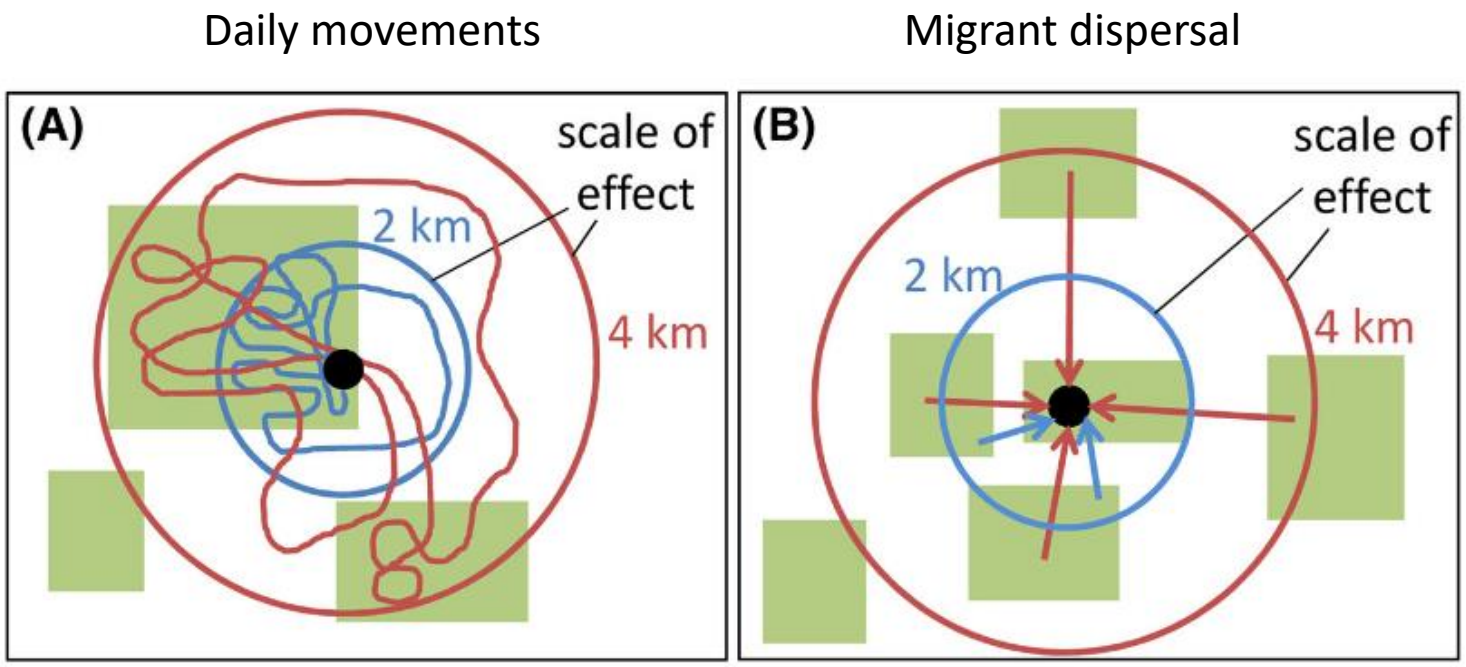
What determines the spatial extent of landscape effects on species?

Paul Miguet · Heather B. Jackson · Nathan D. Jackson · Amanda E. Martin · Lenore Fahrig

Miguet et al. 2016

What drives the (intrinsic) scale of effect?

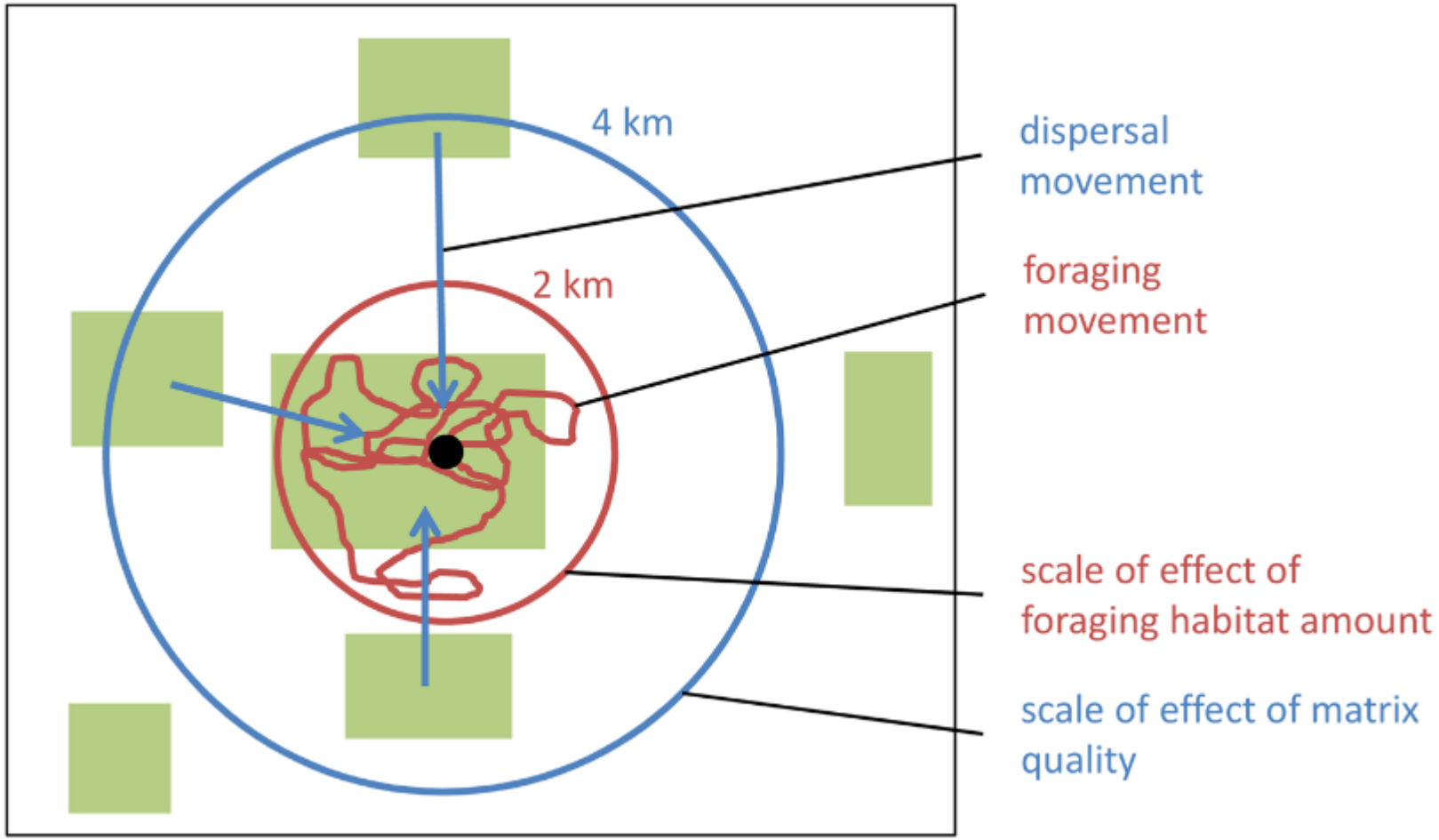
- **Species traits**
  - Dispersal
  - Movement behavior
  - Body size
  - Trophic level
  - Reproductive rates
  - Specialists vs. generalists



Miguet et al. 2016

What drives the (intrinsic) scale of effect?

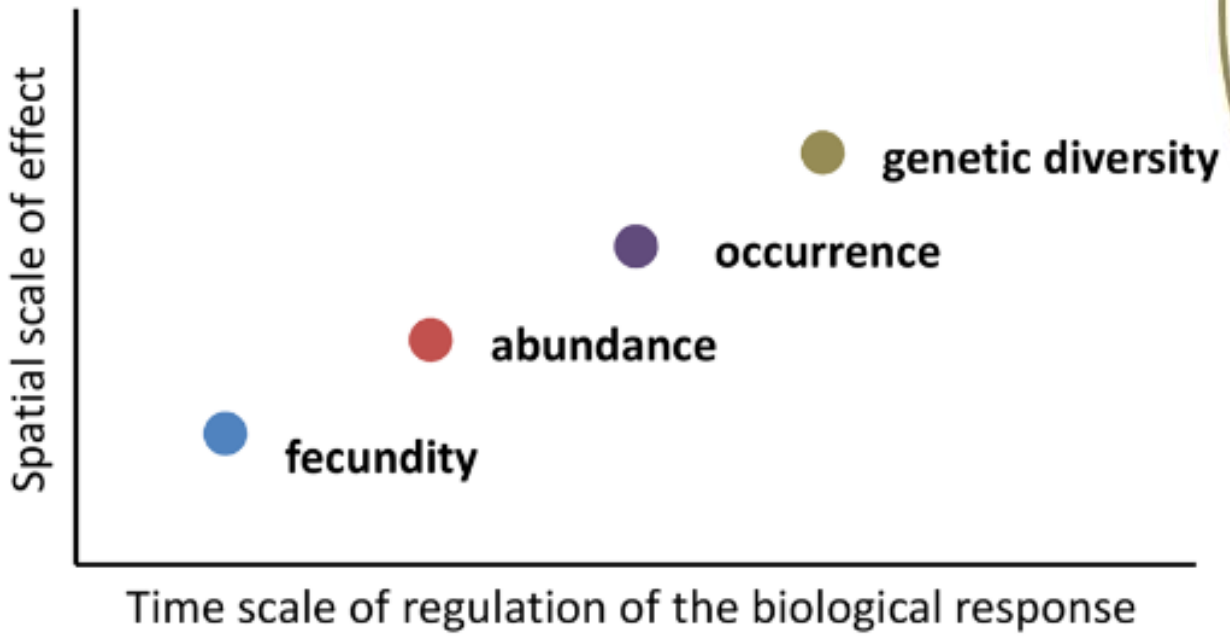
- Species traits
- **Landscape variables**

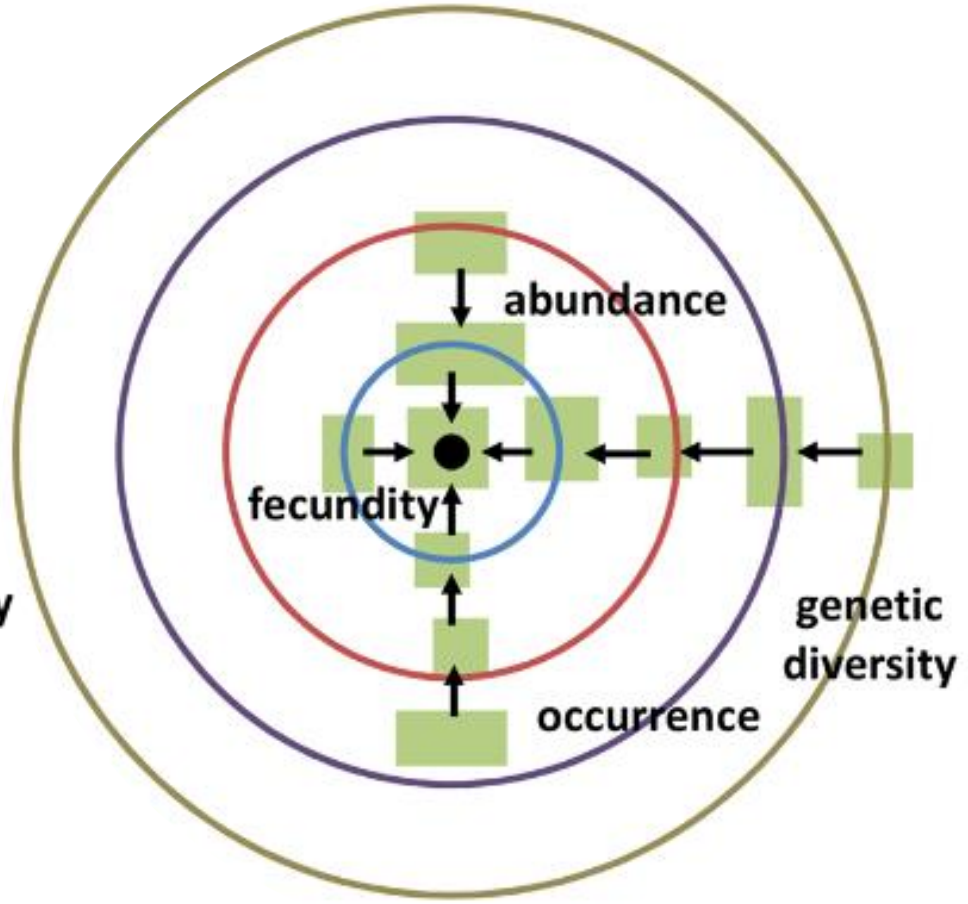


Miguet et al. 2016

What drives the (intrinsic) scale of effect?

- Species traits
- Landscape variables
- **Biological responses**





Miguet et al. 2016

What drives the (intrinsic) scale of effect?

- Species traits
- Landscape variables
- Biological responses
- **Indirect influences on the scale of effect**
  - Abiotic
  - biotic

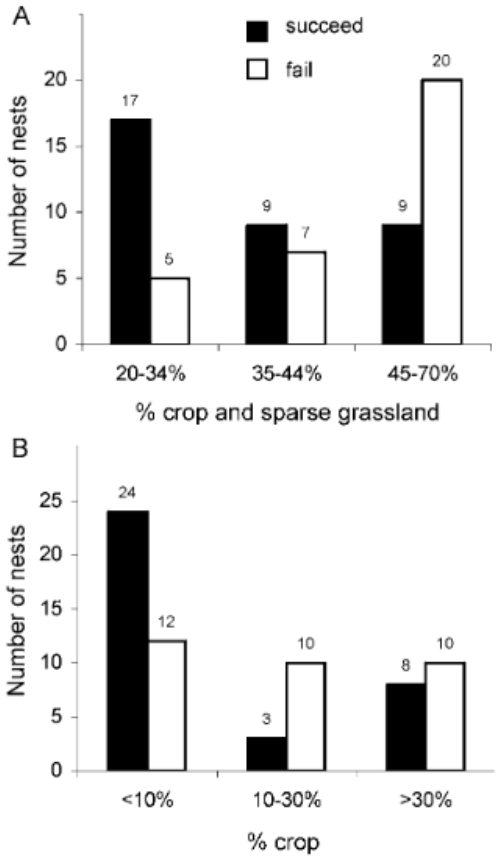


Fig. 1. Sharp-tailed grouse nests ($n = 67$) in landscapes characterized by the proportion of (A) sparse grassland and crop, and (B) crop alone within a 1,600-m extent in Alberta, Canada, 1999–2001. Numbers above bars indicate numbers of nests.

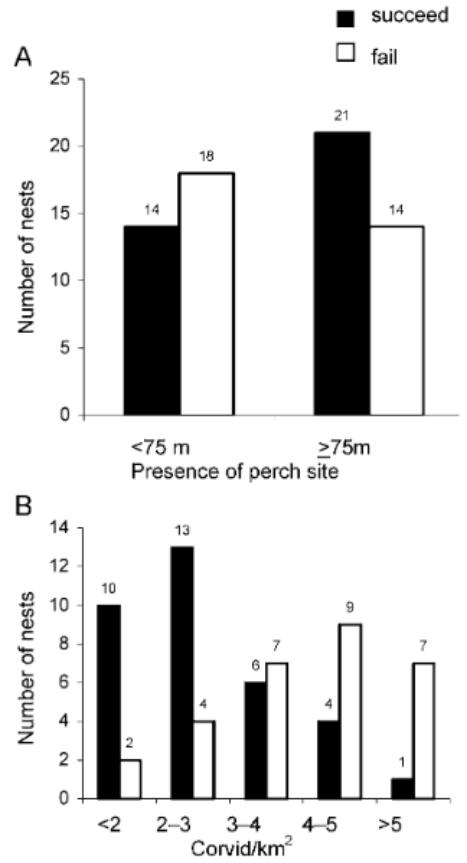


Fig. 2. Sharp-tailed grouse nest success ($n = 67$) in relation to (A) distance from the nearest potential predator perch site, and (B) corvid density measured at the 1,600-m extent in Alberta, Canada, 1999–2001. Numbers above bars indicate numbers of nests.



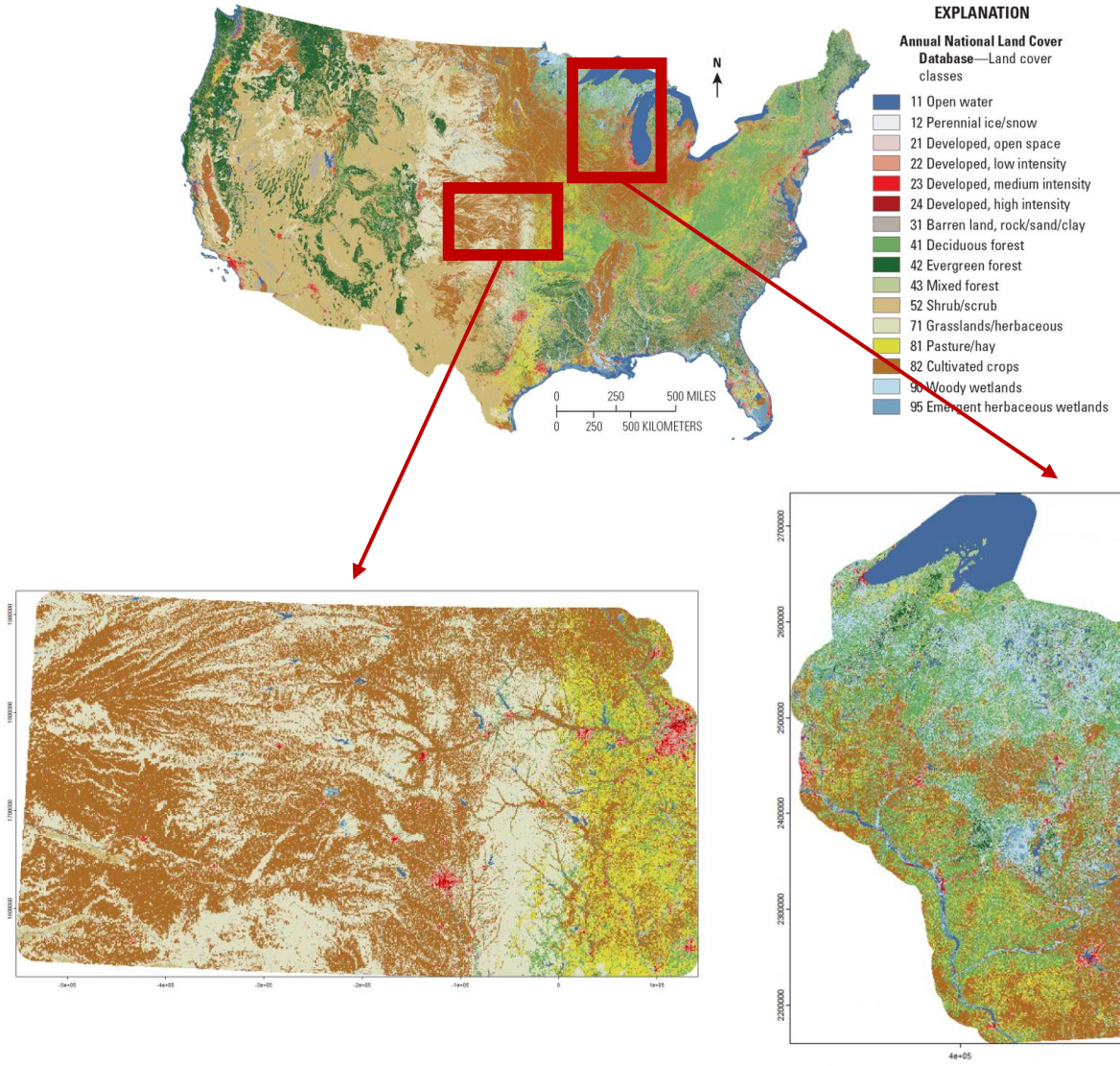
Scale of effect on crop cover for grouse nesting success approx. same as for corvid density.



Manzer & Hannon 2005

What drives the (intrinsic) scale of effect?

- Species traits
- Landscape variables
- Biological responses
- Indirect influences on the scale of effect
- **Regional contexts of a study**



Identifying scale of effect

- Pre-1980's there was not much consideration of scale (scale invariance)
- McArthur, Wiens, Turner, Levin and others really helped bring it to the forefront of ecological thinking during the 1980s-90s
- "…there is no single "correct" scale on which to describe populations or ecosystems…" [Levin 1992]

THE PROBLEM OF PATTERN AND SCALE IN ECOLOGY

THE ROBERT H. MACARTHUR AWARD LECTURE
Presented August 1989
Toronto, Ontario, Canada

by

SIMON A. LEVIN
Department of Ecology and Evolutionary Biology, Princeton University, Princeton, New Jersey 08544-1003 USA, and Section of Ecology and Systematics, Cornell University, Ithaca, New York 14853-2701 USA



Simon A. Levin
MacArthur Award Recipient

Identifying scale of effect

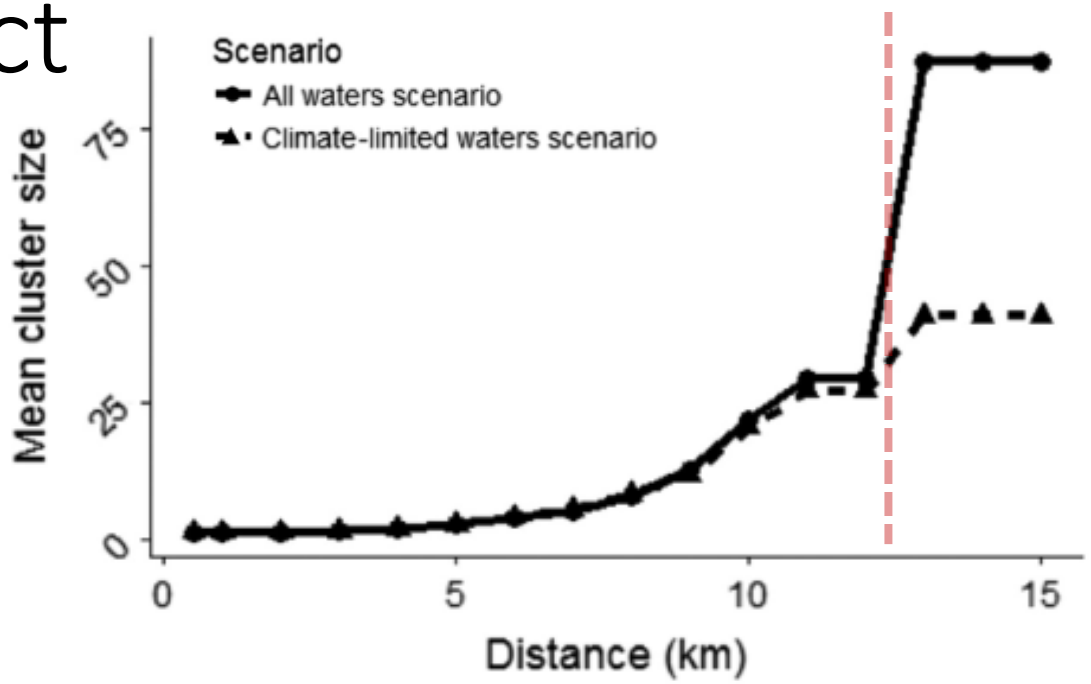
- How do we figure out what scale(s) we should use to collect and analyze data?
- Scales can be dictated by:
  - the motivations a study's research questions
  - practical limitations of data and other factors
  - The philosophy of the approach
    - Patch-mosaic methodology
    - Gradient perspective (sensu McGarigal/Cushman)
- Reviews suggest scales chosen arbitrarily
  - 1993-2007 studies on wildlife habitat research = 70% arbitrary scale selection (Wheatly & Johnson 2009)
  - 2009-2014 studies on habitat selection = 95% not looking at multiple scales (McGarigal et al. 2016)

Identifying scale of effect

- Characteristic scale of effect:
  - A phrase that can have uncertainty around it
  - Sometimes used to describe the dominant pattern/scale for a variable
  - Sometimes used to describe the single scale to examine for all variables

Identifying scale of effect

- Characteristic scale of effect:
  - A phrase that can have uncertainty a
  - Sometimes used to describe the dom
  - Sometimes used to describe the sing

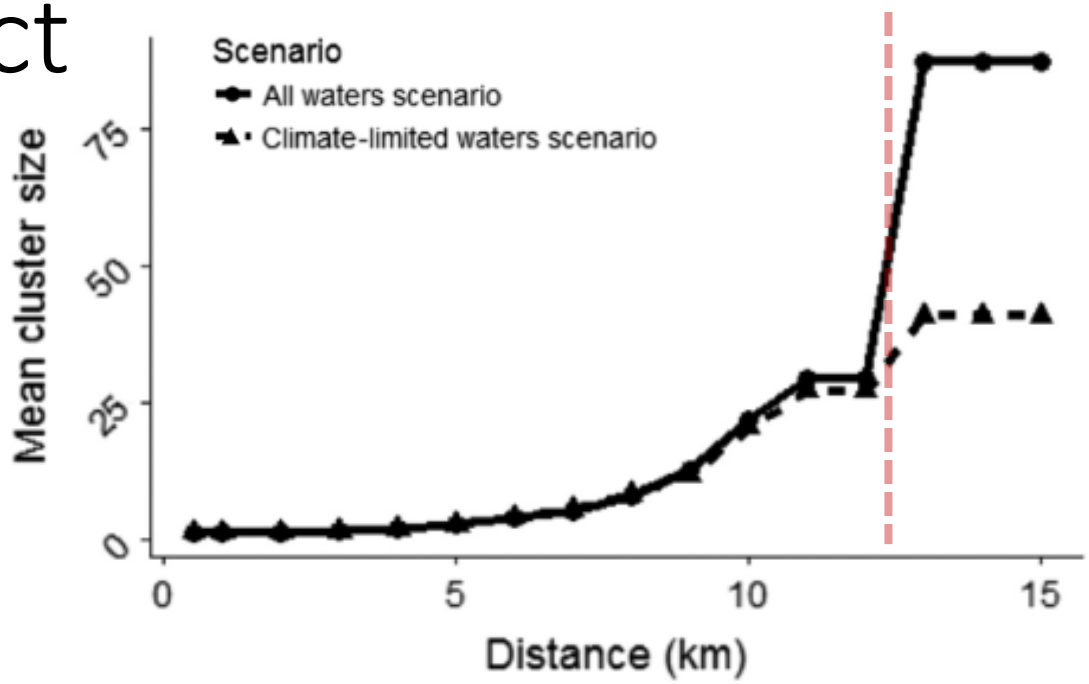


- Critical scales of effect:
  - Often related to some threshold wherein the scale has a crucial impact
  - Conservation planning may require patches within a “critical” dispersal distance

Drake et al. 2017b

Identifying scale of effect

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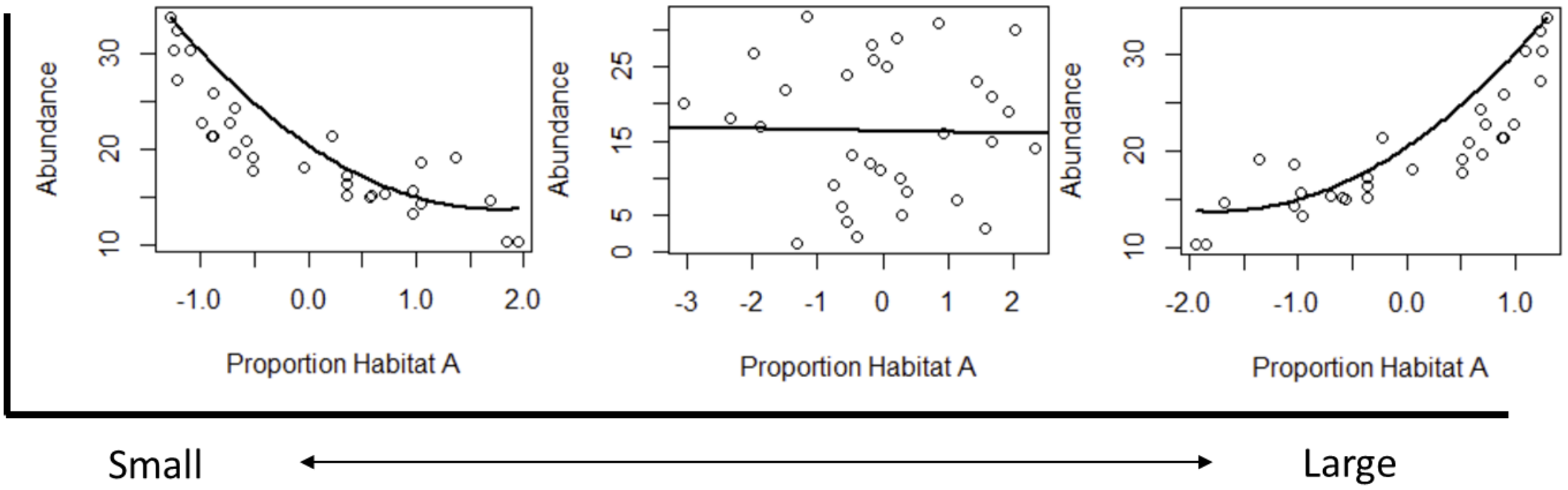
Sometimes characteristic and critical used interchangeably



Drake et al. 2017b

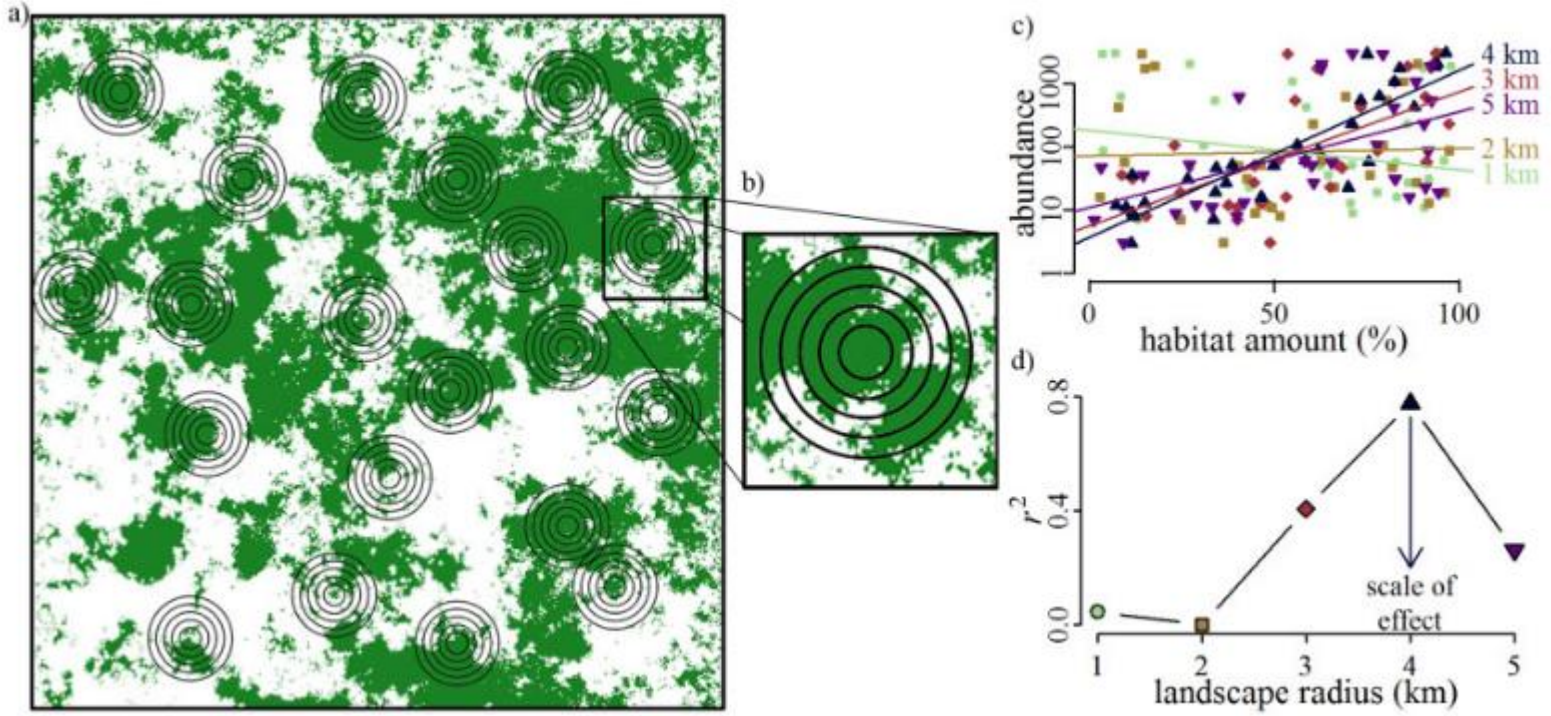
Identifying scale of effect

- The same variable may have an influence across several scales
  - The perceived strength of relationship may change
  - The perceived direction of relationship may change



Buffers

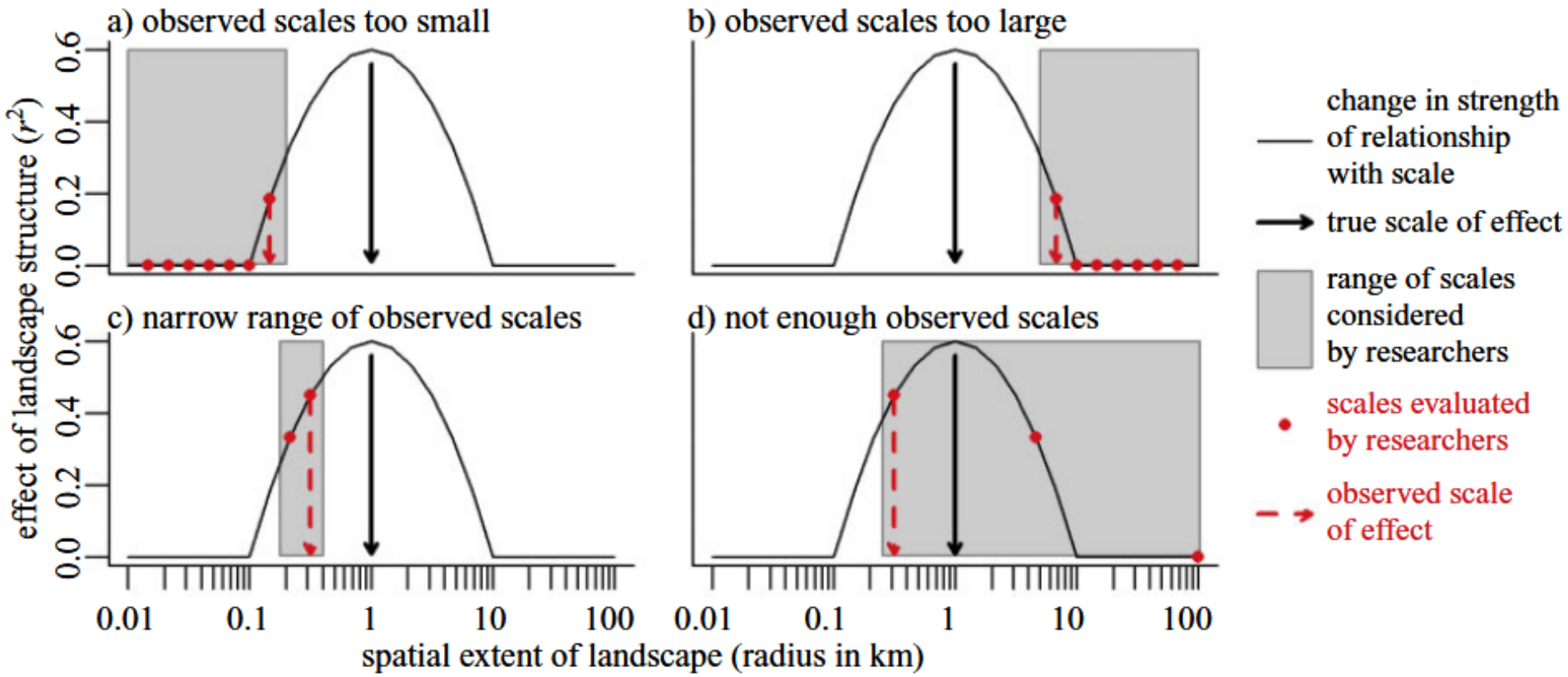
- A common approach is to extract predictor variable amounts at increasing radii around survey points then use model selection to determine which scale is most important for that variable.



Jackson & Fahrig 2015

Buffers

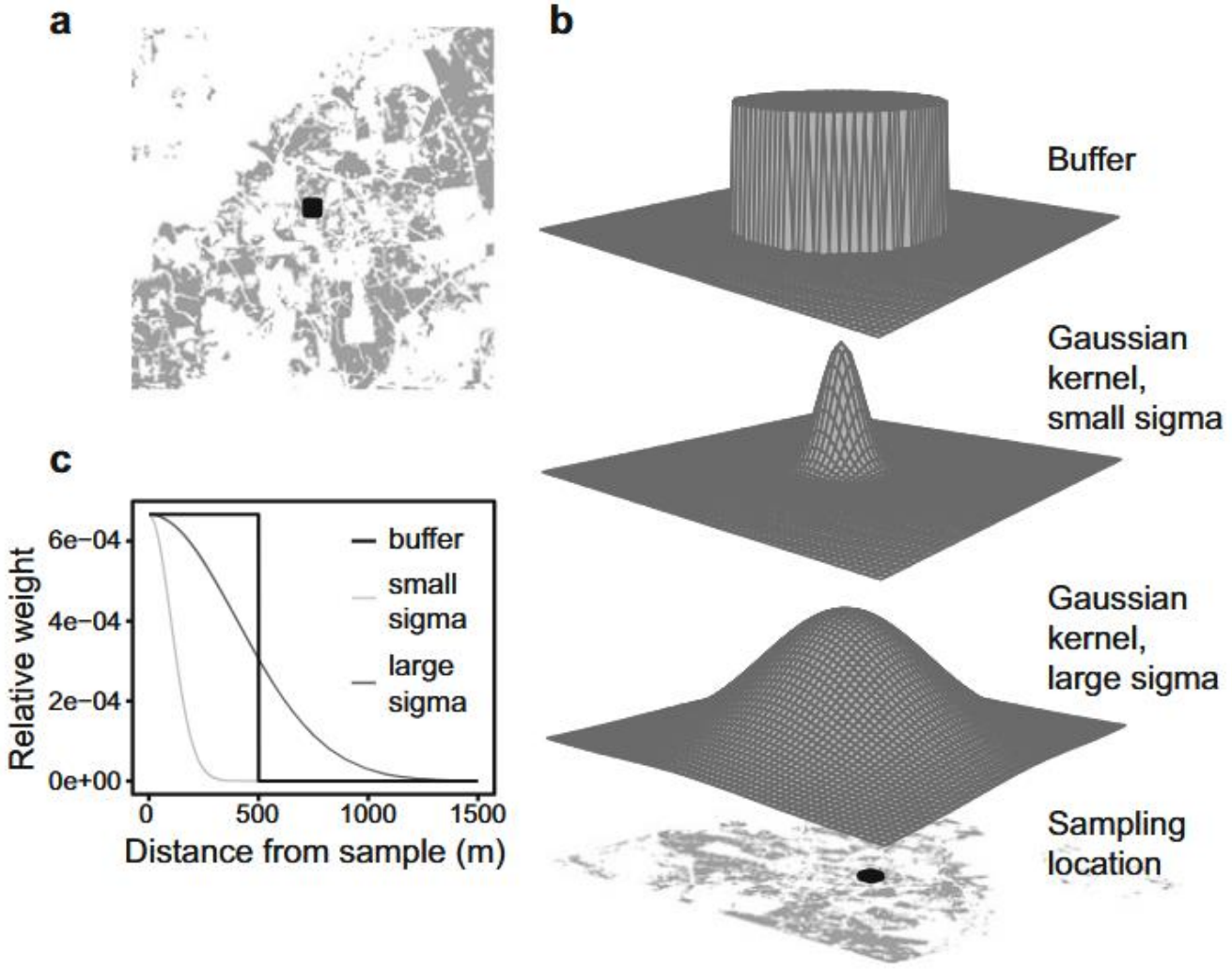
- 44% of studies in a review likely didn't have the correct scale examined



Jackson & Fahrig 2015

Distance weighted effects

- Quantifying contribution of environmental variables as a function of distance
- Kernel functions dictate the shape of decay
- Sigma is the scale parameter

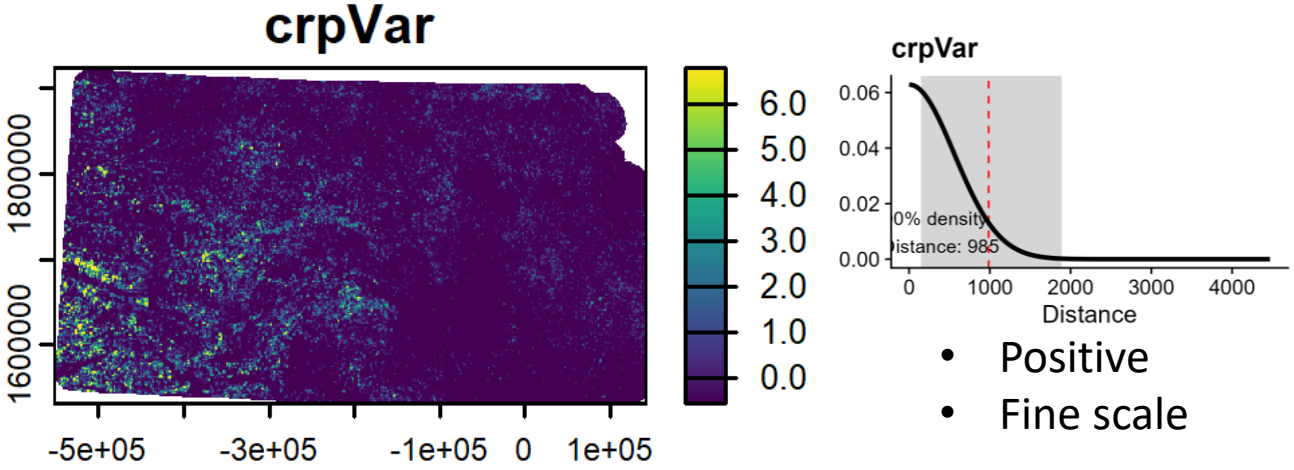


Fletcher & Fortin 2018

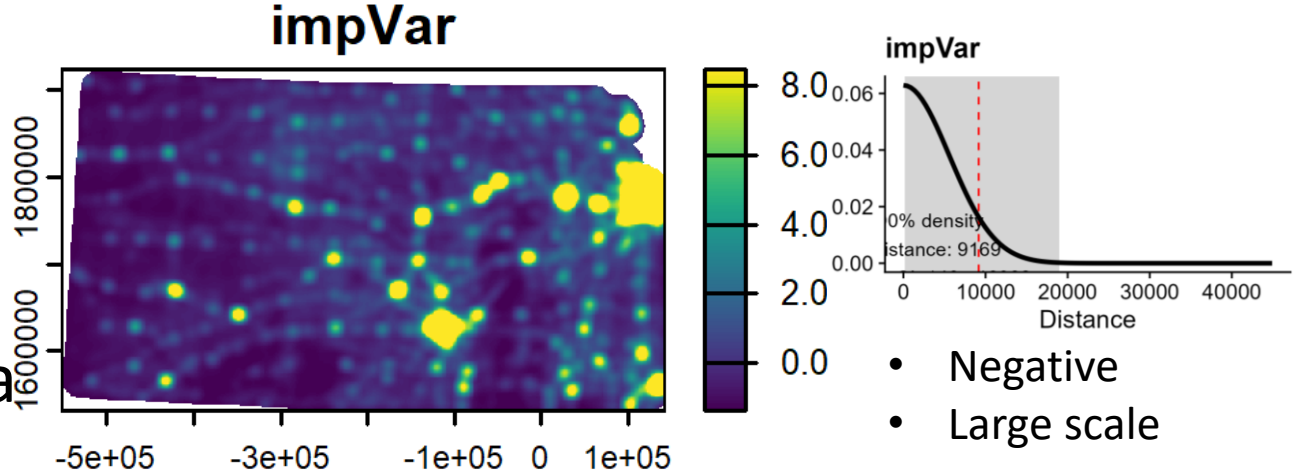
Distance weighted effects



- Quantifying contribution of environmental variables as a function of distance
- Kernel functions dictate the shape of decay
- Sigma is the scale parameter
- The “best” scale will depend on many factors and may differ for each variable
- May have multiple signal optima



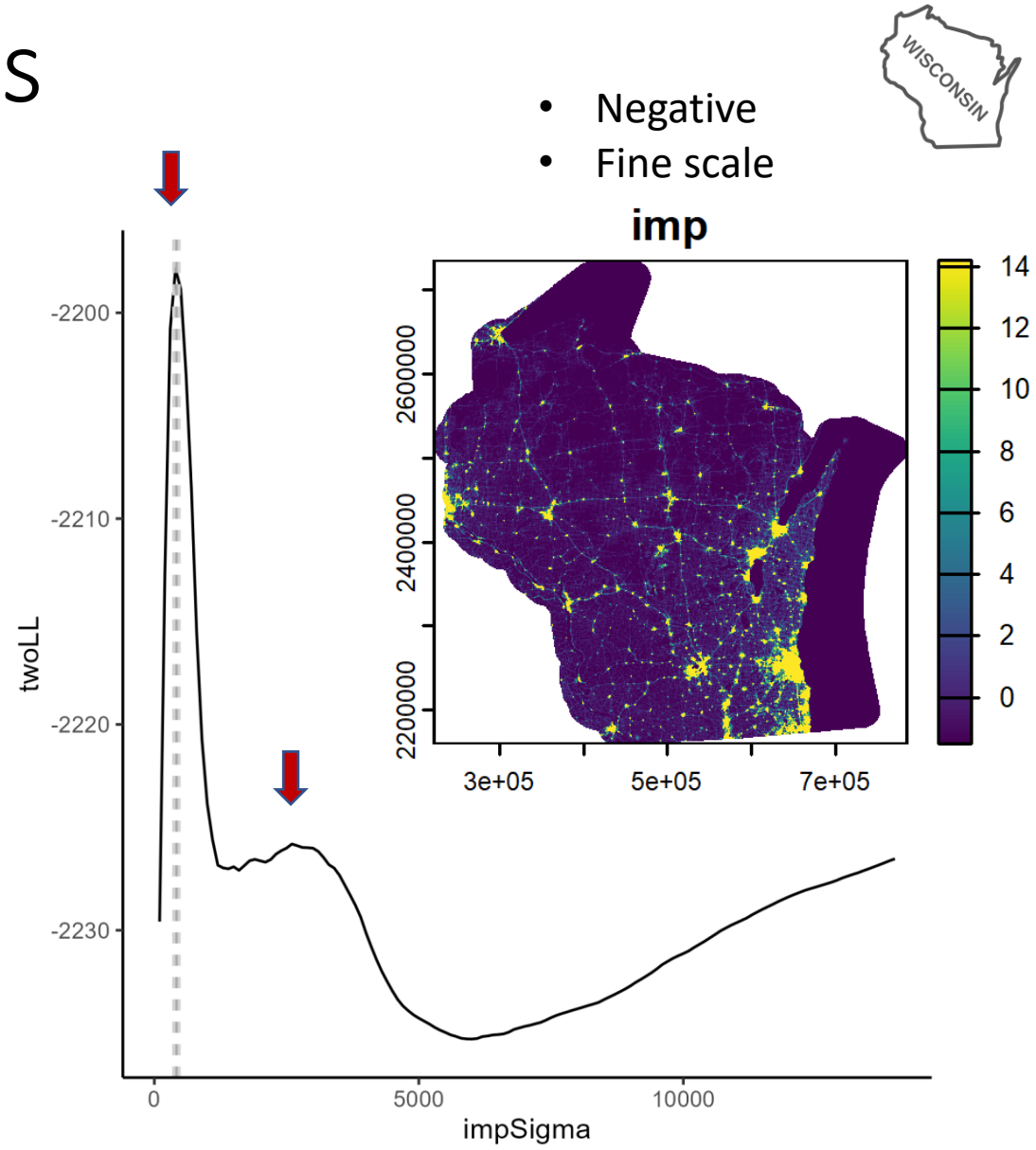
- Positive
- Fine scale



- Negative
- Large scale

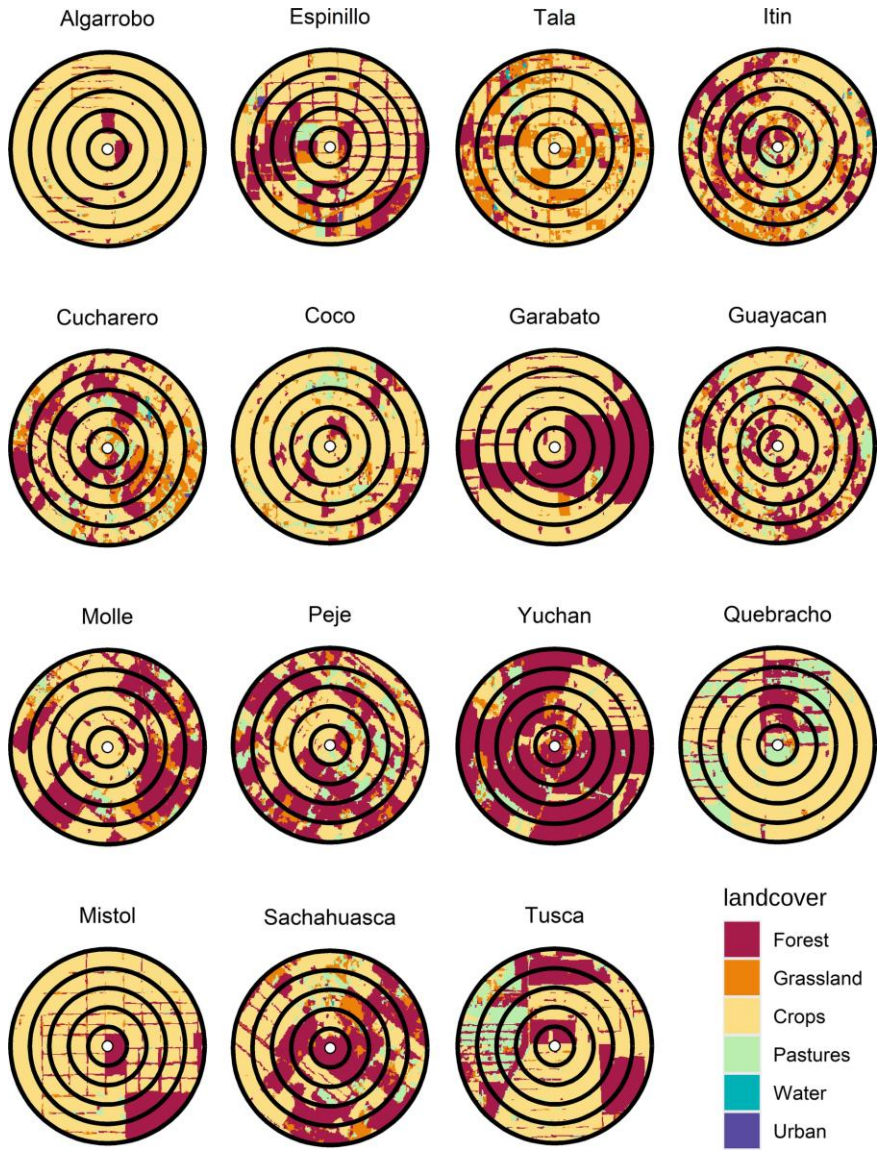
Distance weighted effects

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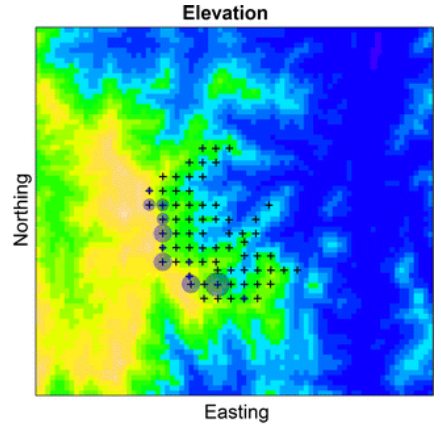
R packages and Methods

- Multilandr: Huais 2024
  - Uses multiple buffer methods
  - Straight forward process
  - Doesn’t have kernel-weighting

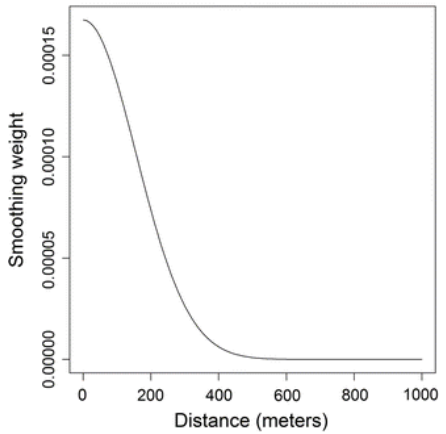


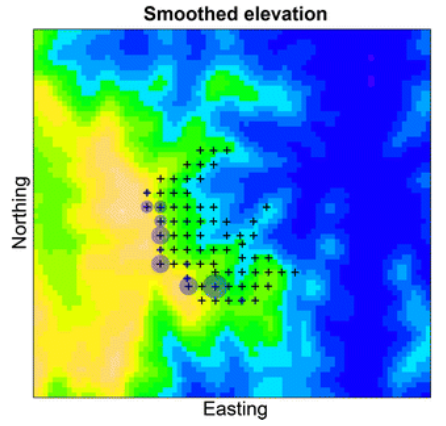
R packages and Methods

- Multilandr: Huais 2024
- KWD Max LL: Chandler & Hepinstall-Cymerman 2016
  - Used smoothing kernels & max likelihood to estimate scale parameters and effects
  - Single covariate analysis



Original

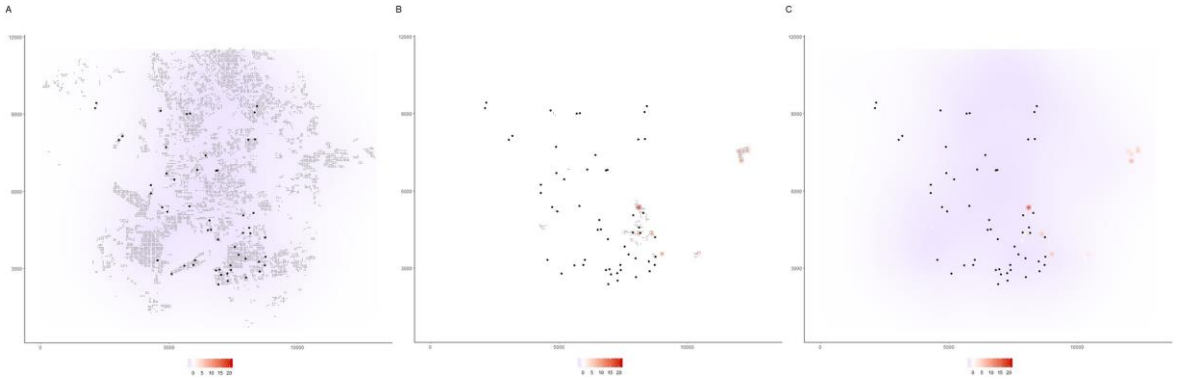


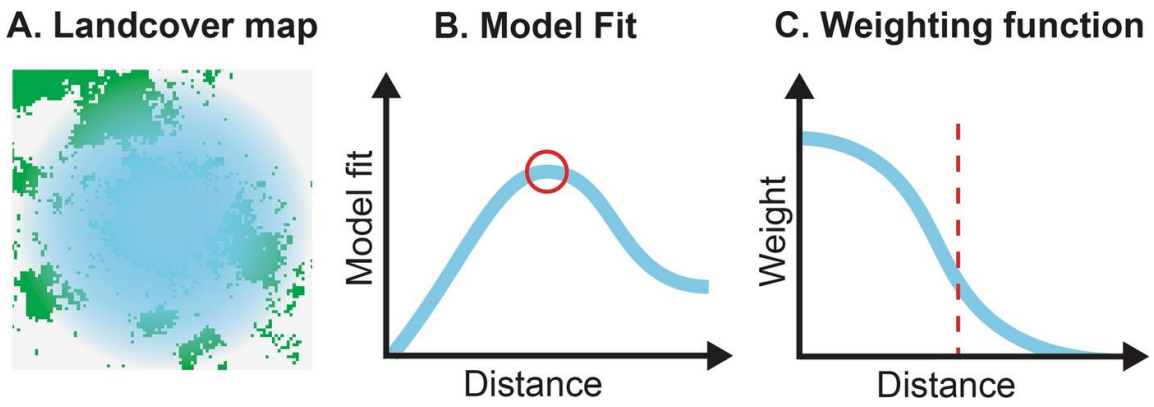


Smoothed

R packages and Methods

- Multilandr: Huais 2024
- KWD Max LL: Chandler & Hepinstall-Cymerman 2016
- Siland: Carpentier & Martin 2021
- Scalescape: Lowe et al. 2022
  - Has a good bootstrapping function

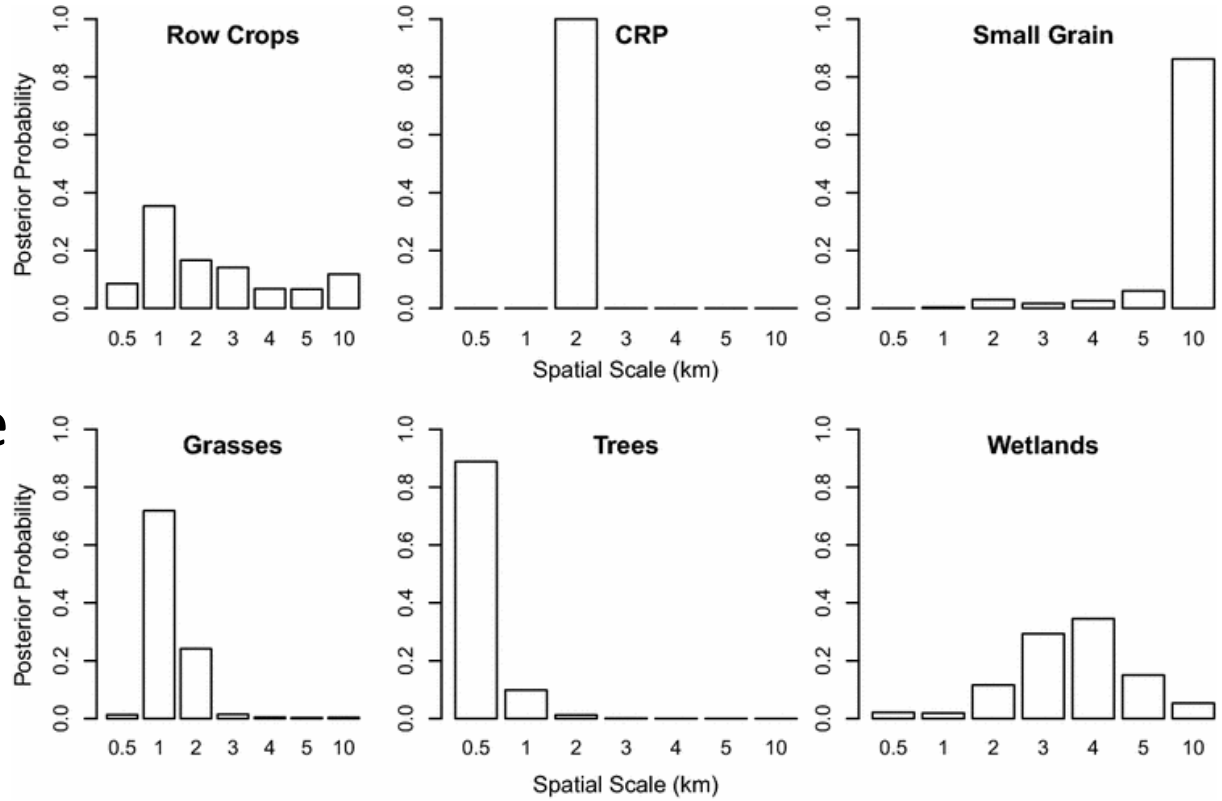




Both Siland and Scalescape offer a good platform for multiscale analyses but have trade-offs for flexibility, interpretability, and tractability.

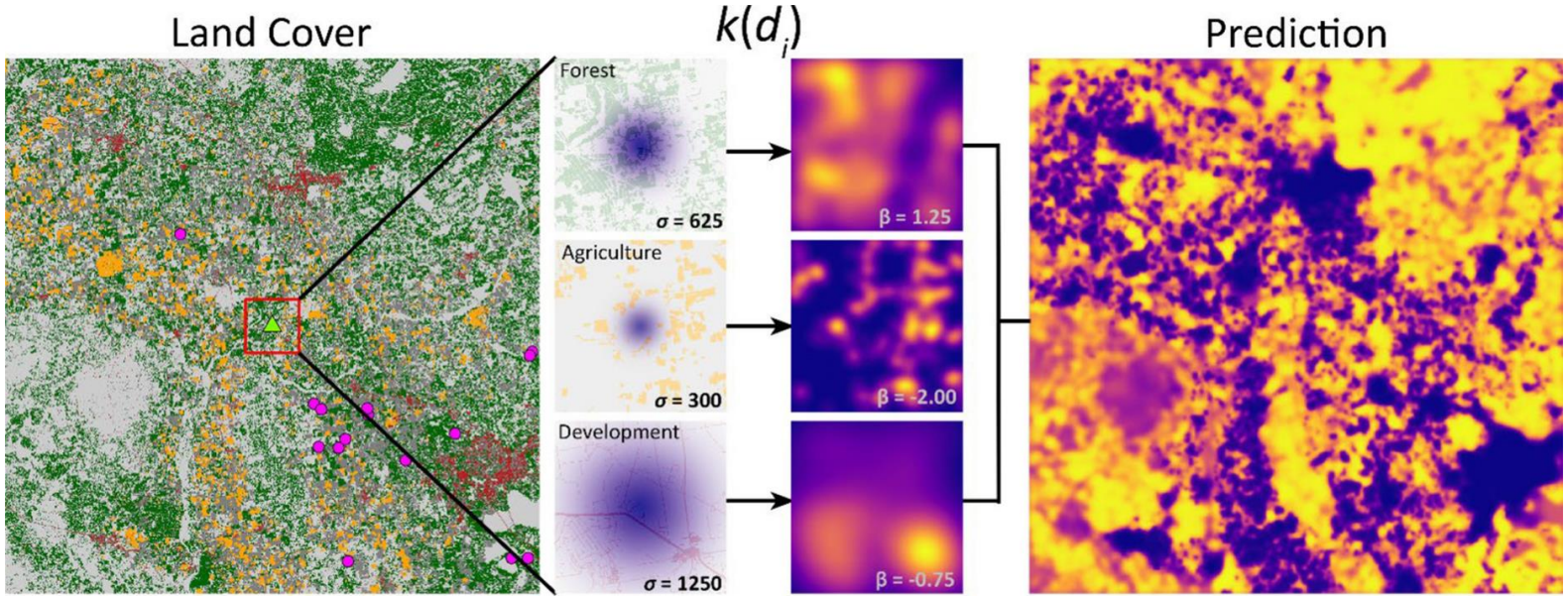
R packages and Methods

- Multilandr: Huais 2024
- KWD Max LL: Chandler & Hepinstall-Cymerman 2016
- Siland: Carpentier & Martin 2021
- Scalescape: Lowe et al. 2022
- BLISS (Bayesian latent indicator scale selection)
  - Stuber et al. 2017
  - Bayesian analytical framework
  - Provides posterior distributions of spatial scales for weight of support
  - Requires fairly specialized/advanced coding skills



multiScaleR: fast, flexible, reproducible

- Expands on these previous methods and packages
- Supports a wide area of model classes including unmarked
- Estimates sigma (the scale of effective distance) directly from the data
- Provides simulation and prediction helper functions

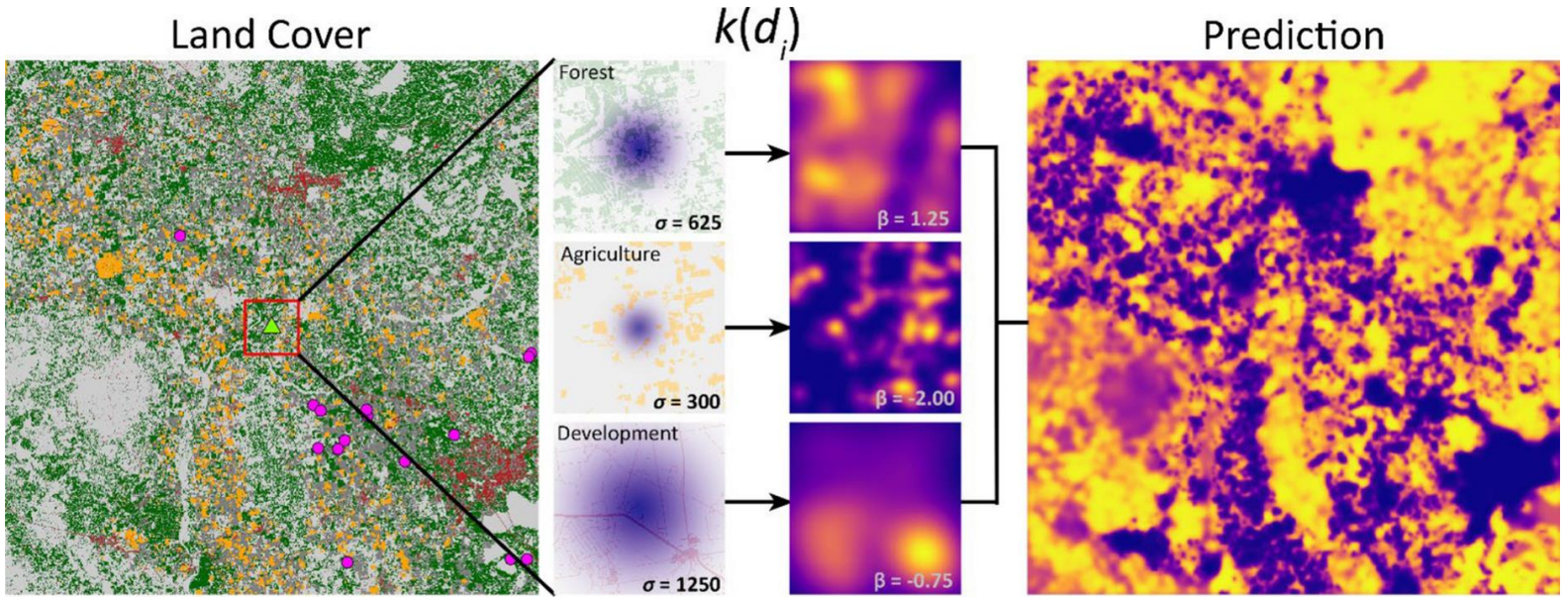


multiScaleR: fast, flexible, reproducible

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A refined approach...



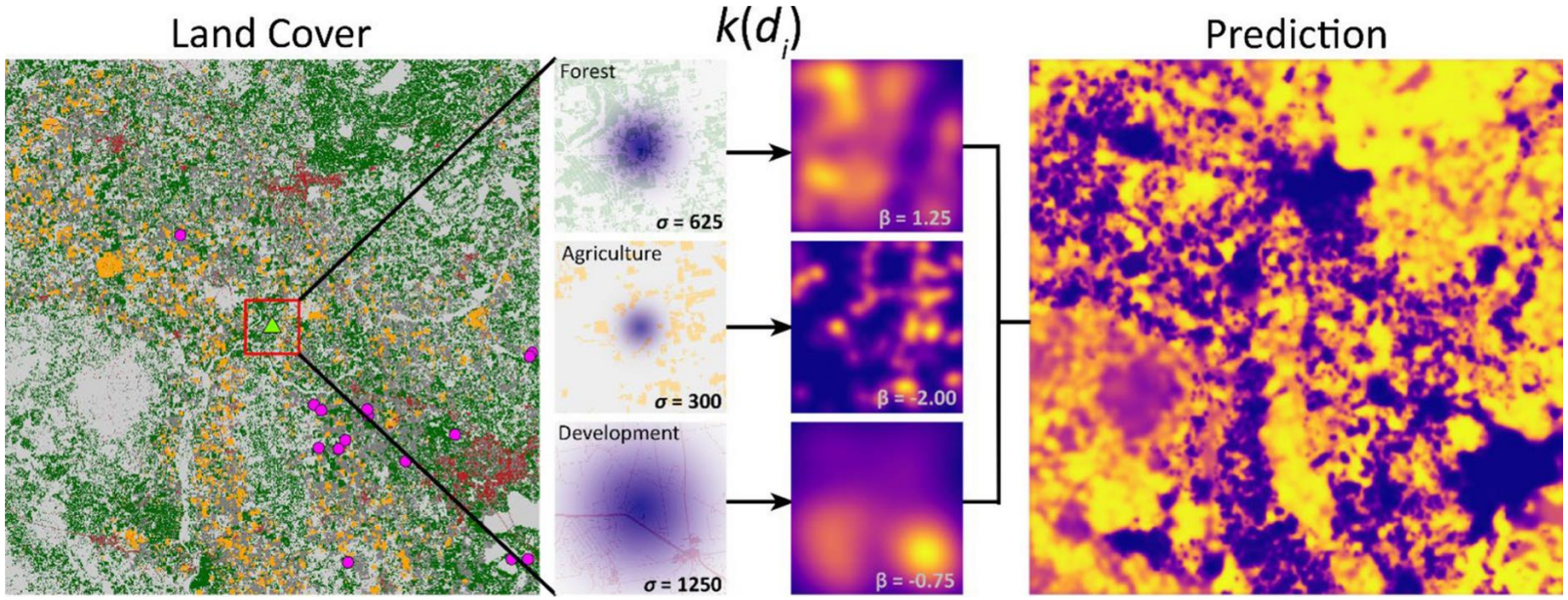


…for refined problems.



2 Estimating it

multiScaleR estimates the scale of effect as a parameter of the model



Extract once

Every raster cell around each site out to max_D, with its distance. Here three layers: forest, agriculture, development.

Weight by distance

A kernel with its own sigma turns each layer into one covariate. Sigma is estimated with the coefficients, by maximum likelihood.

Predict

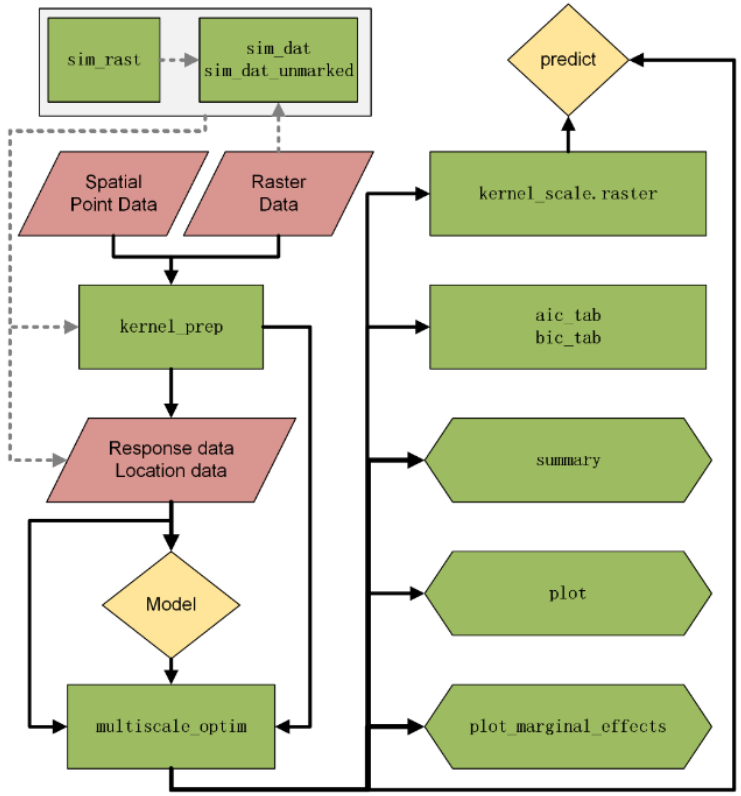
The fitted kernels and coefficients project the model to every cell, so the map is drawn at the scales the data chose.

Sigma comes out of the likelihood, with the coefficients, not from a menu of buffers.

Peterman 2026, Landscape Ecology 41:8, Figure 1. Sampled sites in pink; the focal site and its 5 by 5 km landscape in red.

Six functions carry the whole workflow

kernel_prep()	Extract the cells and distances around each site, once
multiScale_optim()	Estimate sigma and the coefficients together, for any model you pass it
summary(), plot()	Read the fit: sigma, its interval, the two distances, the refit model
profile_sigma()	The likelihood across sigma: the diagnostic that says whether to believe it
aic_tab()	Compare fits on the same sites, every scale parameter counted
kernel_scale.raster()	Project the fitted scales to a map, with terra::predict()

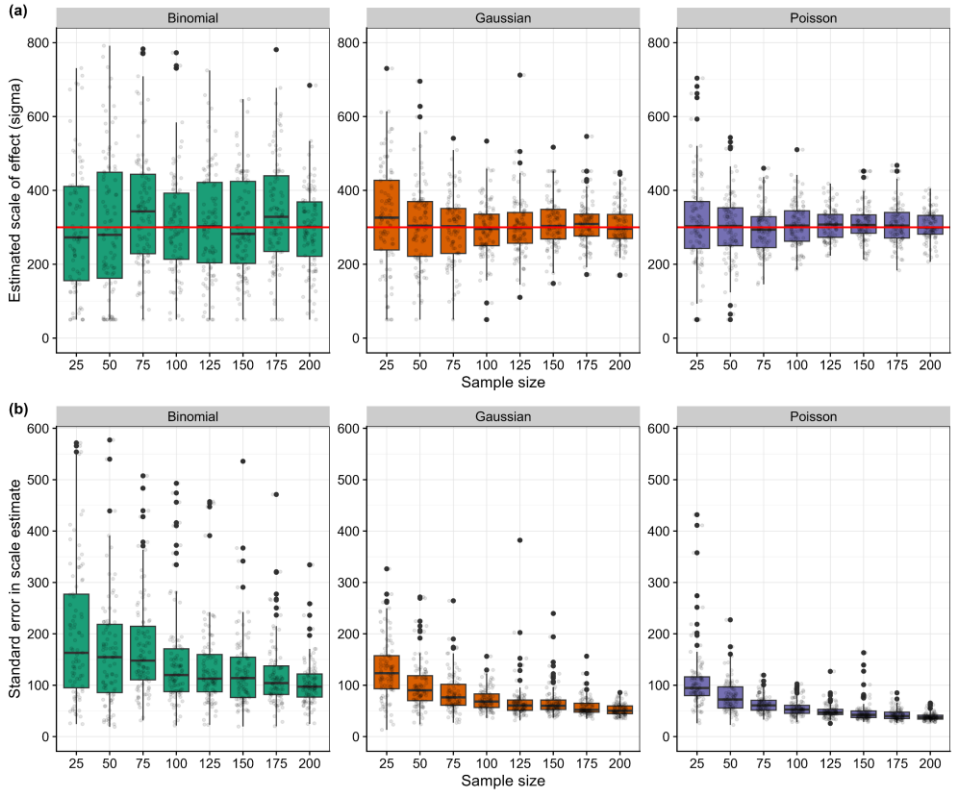


Inputs in red, package functions in green, your model in yellow.

Your model sits in the middle: glm, glm.nb, mixed models, and unmarked occupancy models.

Peterman 2026, Table 1 and the workflow figure. The simulation helpers (sim_rast, sim_dat) are the dotted box.

Simulations recover the true scale, and count and continuous data do it most precisely



What was simulated

A known scale of effect of 300 m (the red line) on a simulated landscape, recovered from 25 to 200 sites with presence-absence, Gaussian, and count responses.

Binomial data are the least precise

A yes or no at each site carries the least information about the surroundings, so the estimates spread widest and their standard errors shrink slowest.

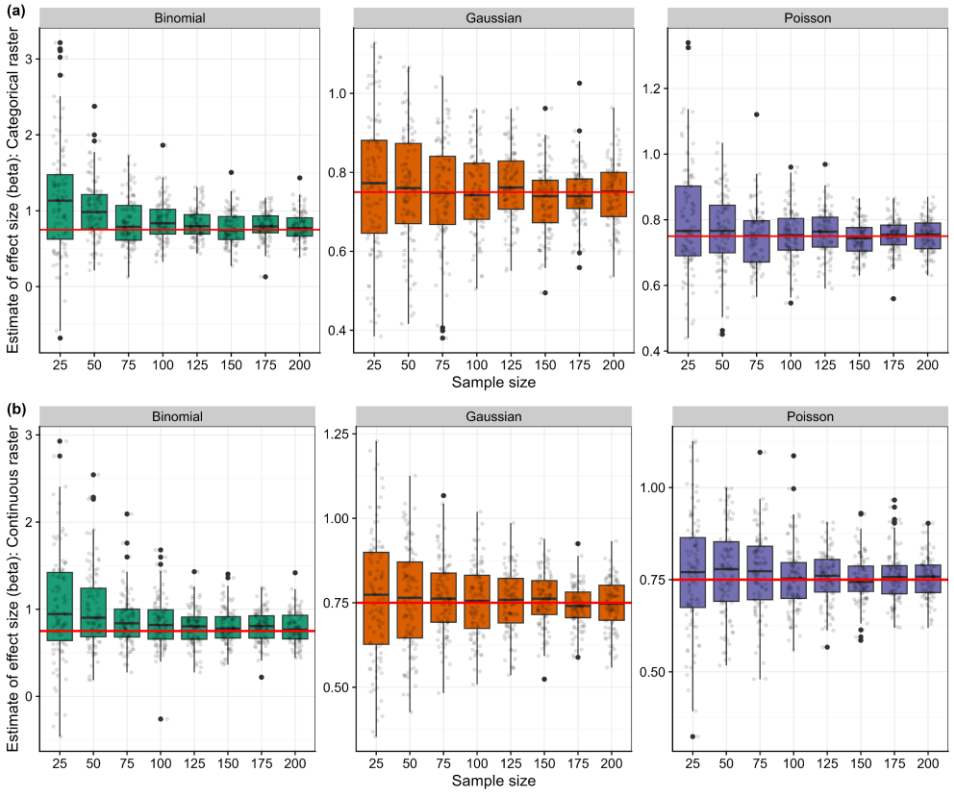
More sites tighten every estimate

For count data the standard error of sigma falls by about two thirds from 25 to 200 sites. The estimates stay centered on the truth throughout.

Unbiased at every sample size; the response type sets how many sites you need.

Peterman 2026, Landscape Ecology 41:8. (a) estimated sigma, (b) its standard error, across replicate simulations.

The coefficients come back too, so the effect is estimated at the right scale



The same simulations, read for beta

The true effect is 0.75 per standard deviation of the kernel-weighted covariate (the red line), on a categorical layer (a) and a continuous layer (b).

Centered from about 75 sites

Gaussian and count responses are centered on the truth throughout; presence-absence runs high at 25 to 50 sites and settles above 75.

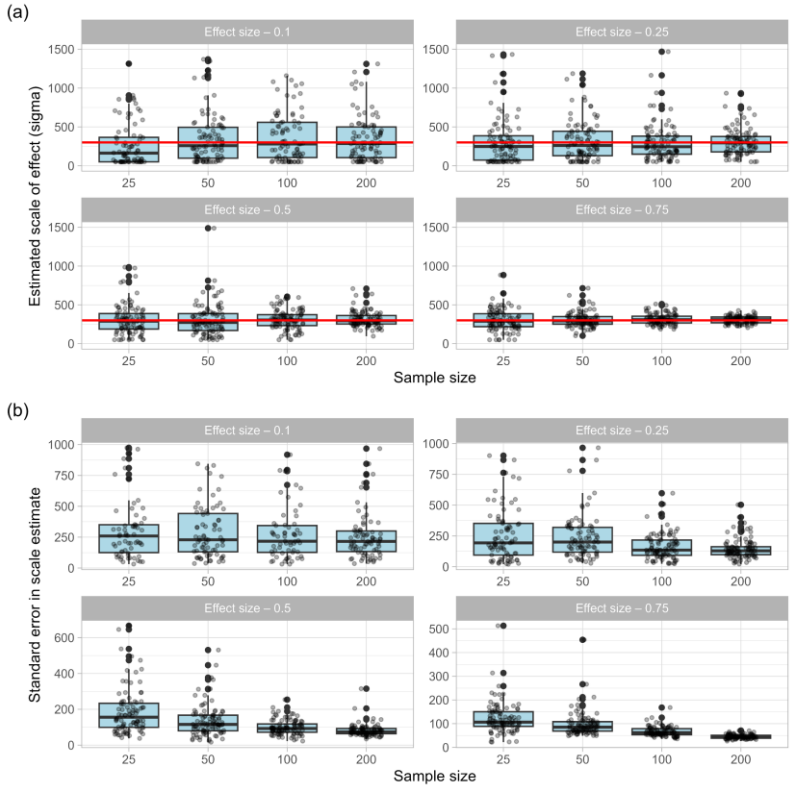
Why it matters

A coefficient estimated at the wrong scale is a coefficient for a different covariate. Estimating scale and effect together is what keeps both honest.

Get the scale wrong and the effect size is wrong with it. Estimate them together.

Peterman 2026, Landscape Ecology 41:8. Effect size on the standardized kernel-weighted covariate.

Weak effects give uncertain scales, and more sites help less than a stronger effect



Effect size 0.1

The scale estimates scatter from near zero to past 1000 m around a true 300 m, and the standard error barely moves between 25 and 200 sites.

Effect size 0.75

Centered and tight from 25 sites on; by 200 sites the standard error is a fraction of what the weak effect allows at any sample size.

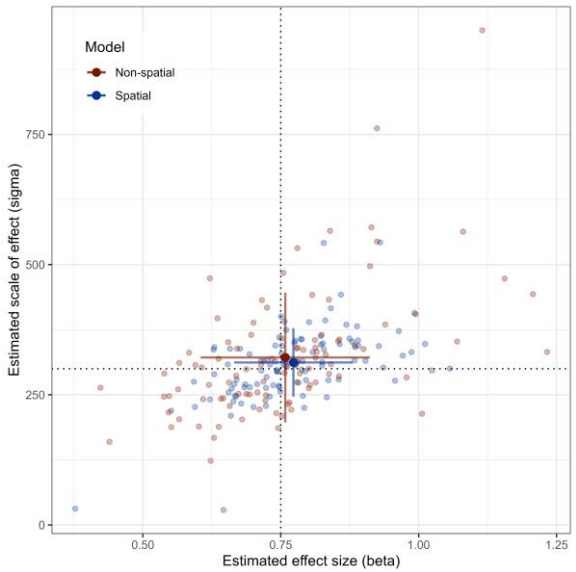
What to conclude in the field

A flat profile and a wide interval can mean the covariate matters little, not that the model failed. Report it as a result. The chipmunk in Block 2 is this case.

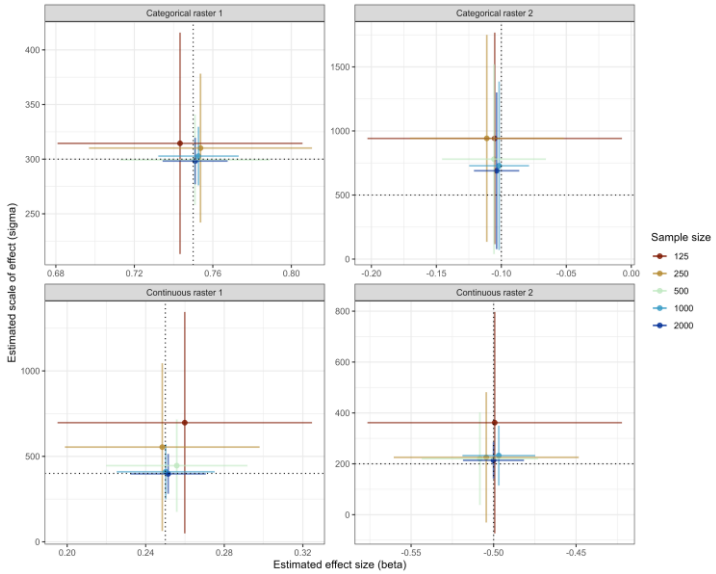
A scale you cannot pin down may be a weak effect, not a bad model.

Peterman 2026, Landscape Ecology 41:8. (a) estimated sigma, (b) its standard error, true sigma 300 m.

Spatial correlation does not break the estimate, and several scales fit at once



(a) Autocorrelated response, two models



(b) Four covariates, four scales, estimated in one model

Autocorrelation in the response

Fit without a spatial term, sigma and beta stay centered on the truth. A spatial model narrows the interval; it does not move the answer.

Several covariates, one likelihood

Two categorical and two continuous layers, each with its own sigma, recovered together. Precision grows with sites.

The broad scale is the hard one

The 500 m categorical covariate needs the most sites. Set max_D with the broadest plausible scale in mind.

Fit every covariate's scale in one model; the data, not the analyst, divide the credit.

Peterman 2026, Landscape Ecology 41:8. Dotted lines mark the true values; bars are 95 percent intervals across simulations.

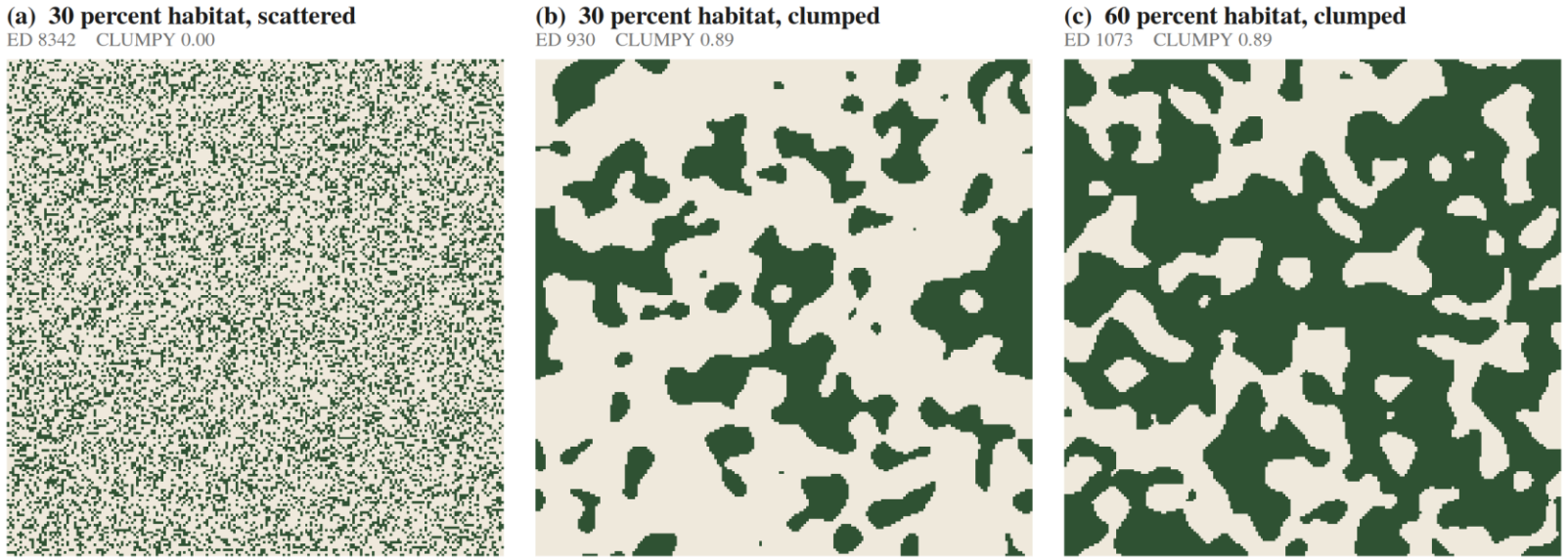
Turn to a neighbor | 2 minutes

Your response: presence-absence, counts, or a continuous measure. How many sites do you have?

Given the simulations, would you trust a scale estimate from your data? What would you report if the profile came back flat?

3 Beyond the mean

A kernel-weighted mean returns the same number for two different landscapes



(a) and (b) share composition

Thirty percent habitat, scattered or clumped. A kernel-weighted mean of habitat is a distance-weighted proportion: it reads 0.30 on both.

(b) and (c) share configuration

The same clumping at 30 and 60 percent habitat. Edge density reads 930 and 1073: nearly one number for maps 30 points apart.

Two questions

How much surrounds a site, and how it is put together. Block 2 asks the first; Block 3 asks the second.

A species can answer either question. The mean only hears the first.

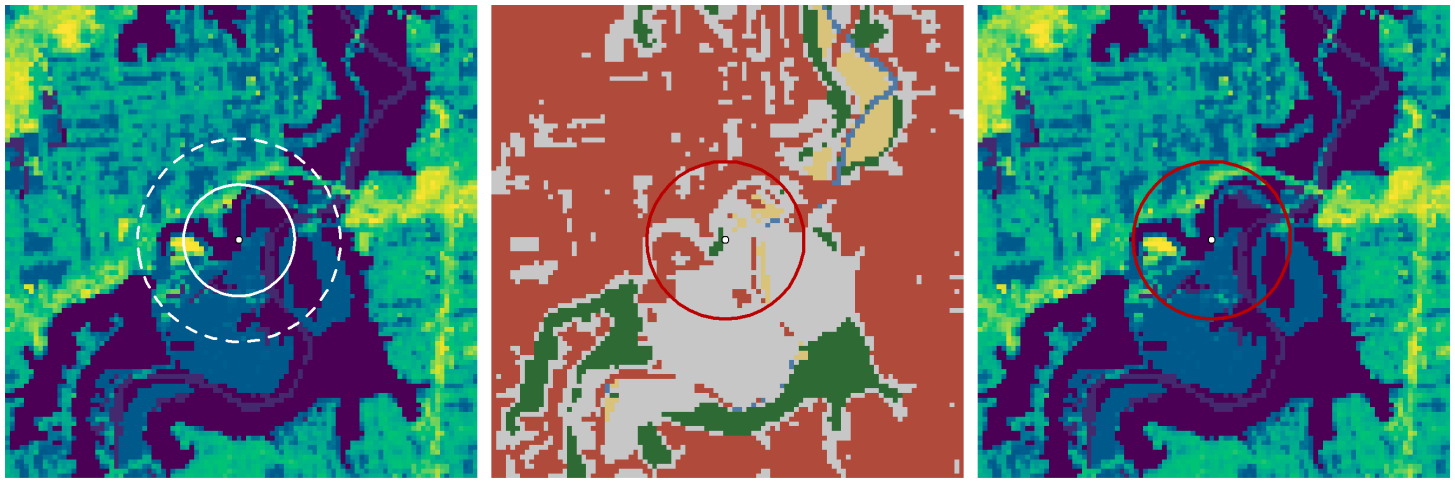
Simulated example from ENR 5374 Lecture 6: binary maps from a thresholded Gaussian random field, 200 by 200 cells; metrics from landscapemetrics.

Three covariate types, each with its own estimated scale

(a) Kernel-weighted mean: kernel_var()
Every cell, weighted by distance; scale: sigma

(b) Landscape metric: landscape_var()
A metric on the classes inside a hard radius

(c) Surface metric: surface_var()
The SD of the index inside the radius: sq = 1.45



Kernel-weighted mean: kernel_var()

A continuous or 0/1 layer: the distance-weighted average of the cells around a site. Scale: sigma, smooth. Block 2.

Landscape metric: landscape_var()

A categorical layer: a FRAGSTATS metric inside a circular buffer. Scale: a hard radius, estimated or fixed. Block 3, opossum.

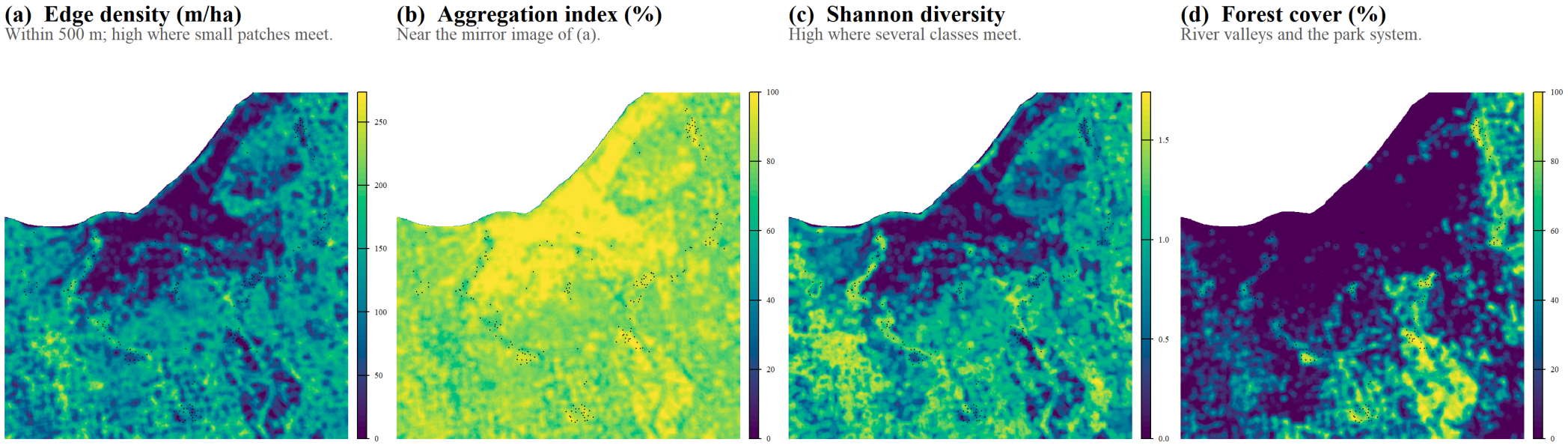
Surface metric: surface_var()

A continuous layer: roughness, skew, or slope. Scale: a hard radius, or sigma with weighted = TRUE. Block 3, deer.

They go in one model, and multiScale_optim() estimates every scale at once.

One 3 km window of the Cleveland data (Moll et al. 2020): the PCA urbanization index and the six-class land cover. Peterman 2026; multiScaleR 0.7.2, msr_vars().

A categorical map yields many covariates, and most of them say the same thing

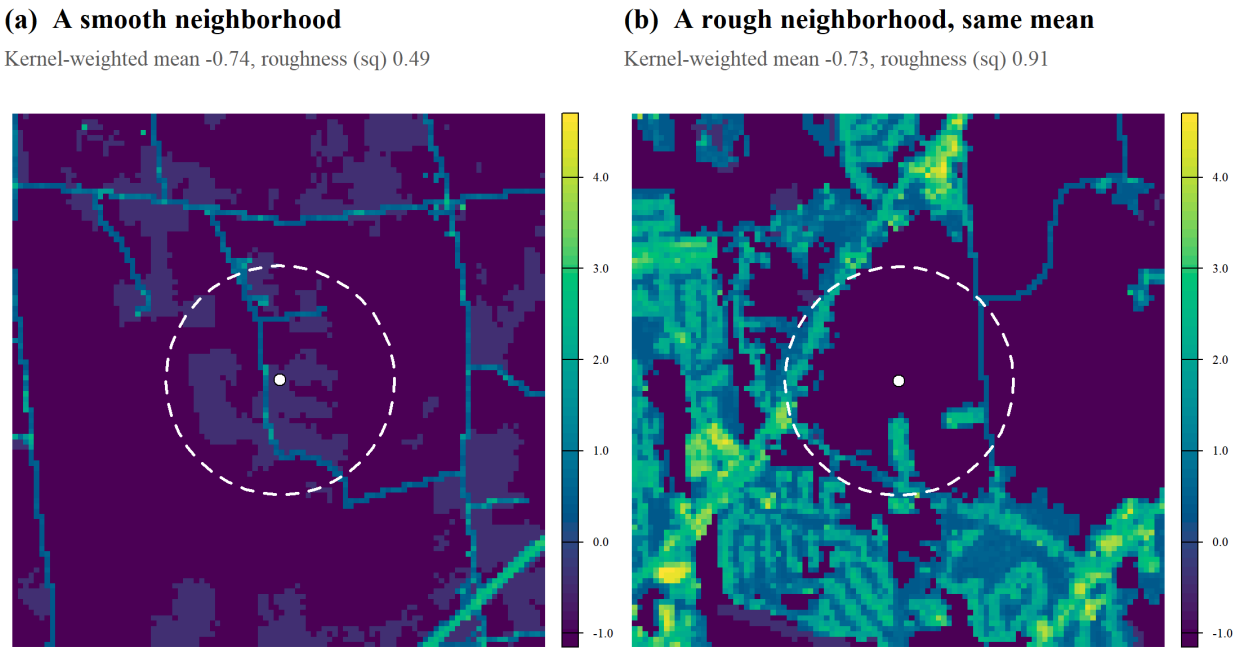


Twenty-five metrics in five families, all to the FRAGSTATS definitions in landscapemetrics. At the 207 cameras, edge density and aggregation index correlate at -1.00, and Shannon diversity and marginal entropy at 1.00.

One metric per idea: amount, edge, clumping, diversity. Look at the maps before you fit.

Land cover: 2016 Annual NLCD in six classes (Block 1); the Block 3 metric maps at a 500 m radius. Hesselbarth et al. 2019; Nowosad and Stepinski 2019; Cushman et al. 2008.

Surface metrics measure how variable a surface is, not how high



Two cameras, 3 km windows of the PCA index. Dashed circle: the 90 percent radius of a 300 m kernel.

Distribution metrics

sa average roughness, sq RMS roughness (the standard deviation), ssk skewness, sku kurtosis. These accept weighted = TRUE: the kernel that weights the mean weights the deviations, and sigma is estimated with a smooth likelihood.

Slope metrics

sdq RMS slope and sdr surface area ratio. Hard radius only; they read terrain or structural ruggedness.

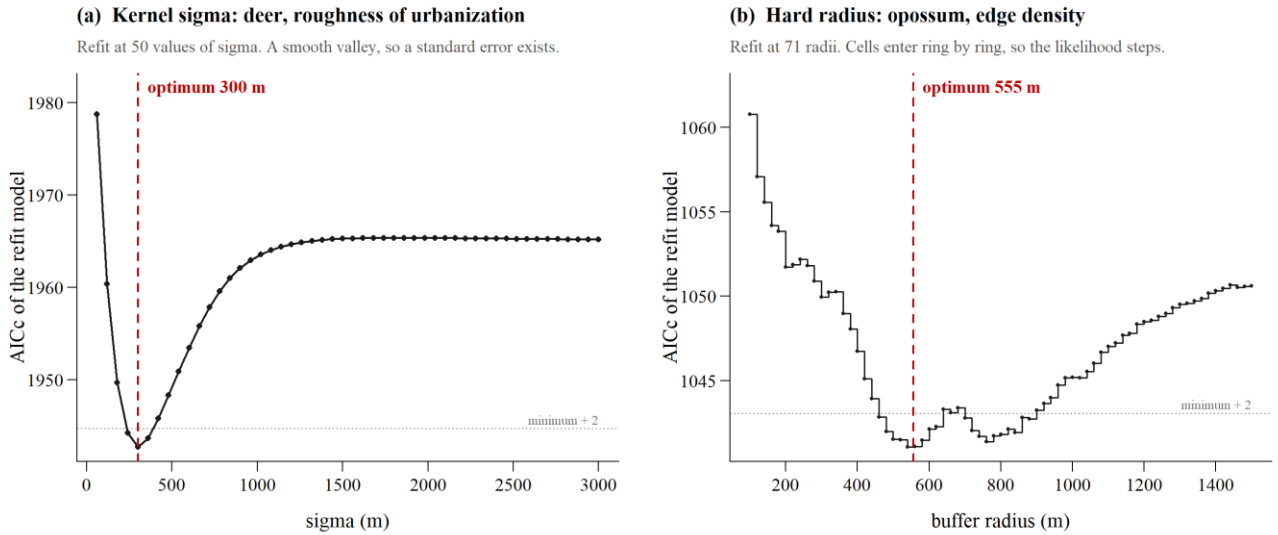
Same mean, different landscape

A neighborhood with a moderate mean can be a uniform suburb or a mosaic of forest and dense development. Here the means agree to within 0.01 and the roughness differs 1.9-fold.

Deer in Block 3 track where development and forest interleave, not how developed the surroundings are on average.

McGarigal et al. 2009; definitions follow the geodiv package; data from Moll et al. 2020.

A kernel sigma and a buffer radius are two currencies, and only one has a standard error



Kernel sigma

A smooth likelihood: a Wald standard error and a profile interval. Report the 2D radius, 2.15 sigma, the circle holding 90 percent of the weight; the 1D distance (1.65 sigma) holds only 74 percent.

Hard radius

Cells enter the buffer ring by ring, so the likelihood moves in steps. A Hessian-based standard error assumes a slope; a staircase has none. Report the profile interval.

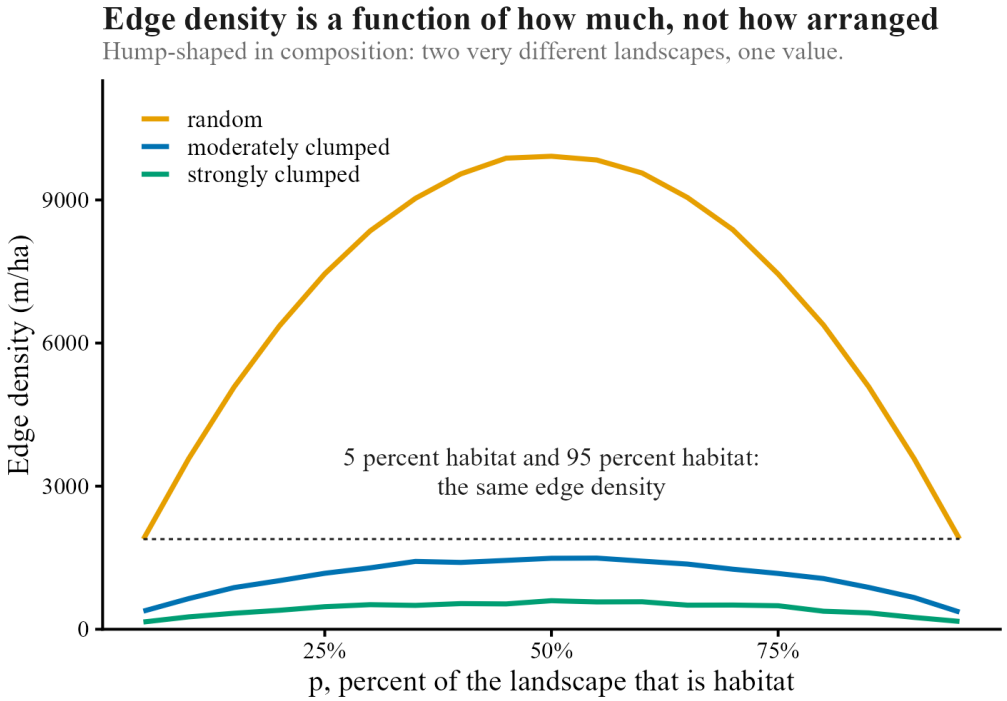
Same neighborhood, two units

Deer roughness: sigma 300 m, 2D radius 644 m; the hard-radius version 715 m. Opossum edge density: radius 555 m, profile 437 to 958 m.

Put every scale in one table in one currency: a planar radius, with a profile interval.

Refits of the Block 3 deer roughness and opossum edge-density models on multiScaleR 0.7.2. Moll et al. 2020 report the 2D radius as scale90.

Configuration metrics move with habitat amount, so fit amount and configuration together



Simulated binary maps, three levels of clumping, habitat amount swept from 5 to 95 percent.

The hump

Edge density is low when there is almost nothing to have an edge, high in the middle, low again when the class fills the map. One value, two compositions.

The usual fix does not work

Putting percent habitat in the model as a linear covariate does not remove a nonlinear dependence (Urban 2023, page 132).

What an estimated scale changes

Amount and configuration enter one model, each at its own estimated scale, after a correlation screen, and AICc compares them on the same sites. The question becomes which one carries the signal once the other is in.

One metric per idea, each at its own scale, and the AICc table decides.

Simulation from ENR 5374 Lecture 6 (same seed and sweep), redrawn for this slide; Urban 2023, page 132; Turner and Gardner 2015, Figure 4.10.

The workflow does not change: name the covariates, then prepare, fit, optimize, read

1. **Define**
   msr_vars() with one landscape_var() or surface_var() per idea. Leave radius out to estimate it.

2. **Prepare**
   kernel_prep() with scale_vars. Keep the cell data (the default): metrics need every cell. Lean mode is for kernel means only.

3. **Fit and optimize**
   The same glm.nb() and multiScale_optim() as Block 2. Screen cor(ki$kernel_dat) first.

4. **Read**
   summary(opt, profile = TRUE), diagnostics(opt)$sample_size, aic_tab(), and kernel_scale.raster() for metric maps with or without a model.

```
vars <- msr_vars(
  forest = landscape_var("landcover", metric = "pland",
                         class = 4),
  edge   = landscape_var("landcover", metric = "ed"),
  rough  = surface_var("urb_pca", metric = "sq",
                       weighted = TRUE))

ki  <- kernel_prep(cam_sf, c(landcover, urb),
                   max_D = 2000, scale_vars = vars)
round(cor(ki$kernel_dat), 2)

dat <- data.frame(visits  = site_data$opossum,
                  n_weeks = site_data$n_weeks,
                  ki$kernel_dat)
mod <- glm.nb(visits ~ edge + offset(log(n_weeks)),
              data = dat)
opt <- multiScale_optim(mod, ki)
summary(opt, profile = TRUE)
aic_tab(list(opt_pca, opt),
        mod_names = c("PCA mean", "Edge density"))
```

Three new functions, no new workflow.

vignette("landscape_metric_covariates") and vignette("surface_metric_covariates"), multiScaleR 0.7.2.

Turn to a neighbor | 2 minutes

For a species you work on, is "how much" or "how mixed" the better hypothesis?

Which layer would carry it? And would you expect its scale to be smaller or larger than the scale for the mean?

References and other works

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